

BFS Traversal

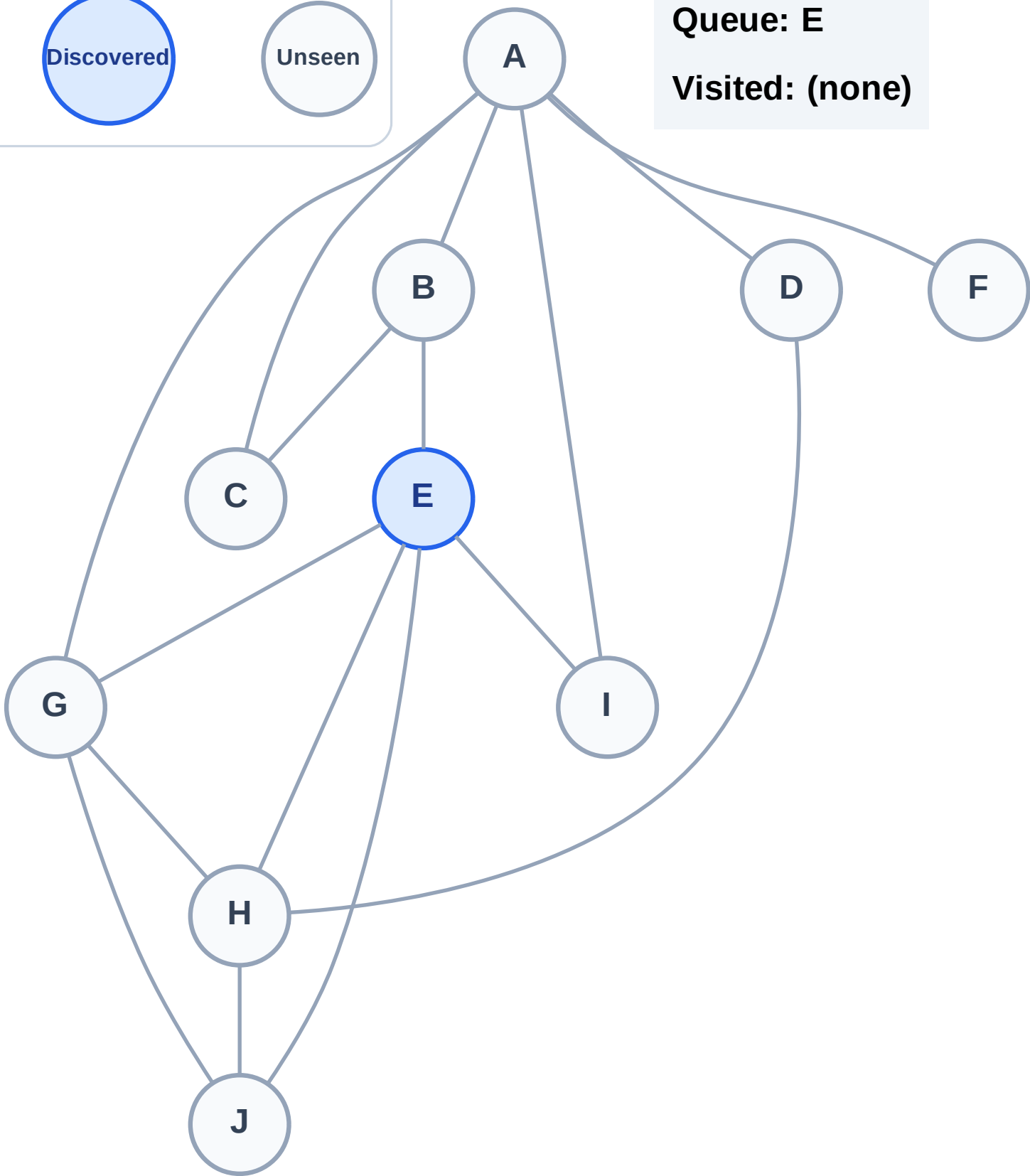
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

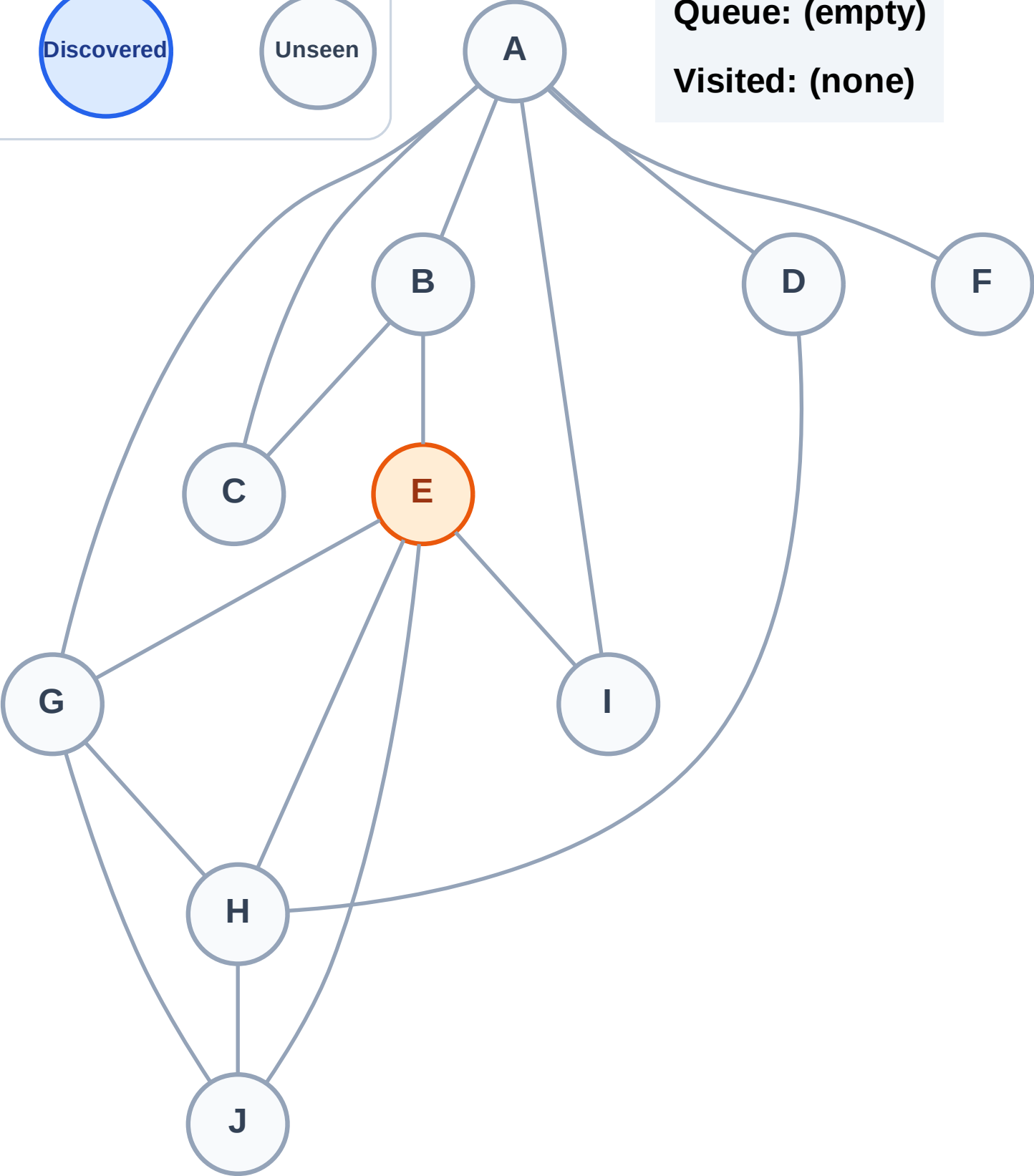
Complete

Processing

Discovered

Unseen

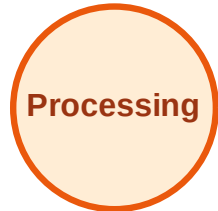
Queue: (empty)
Visited: (none)



BFS Traversal

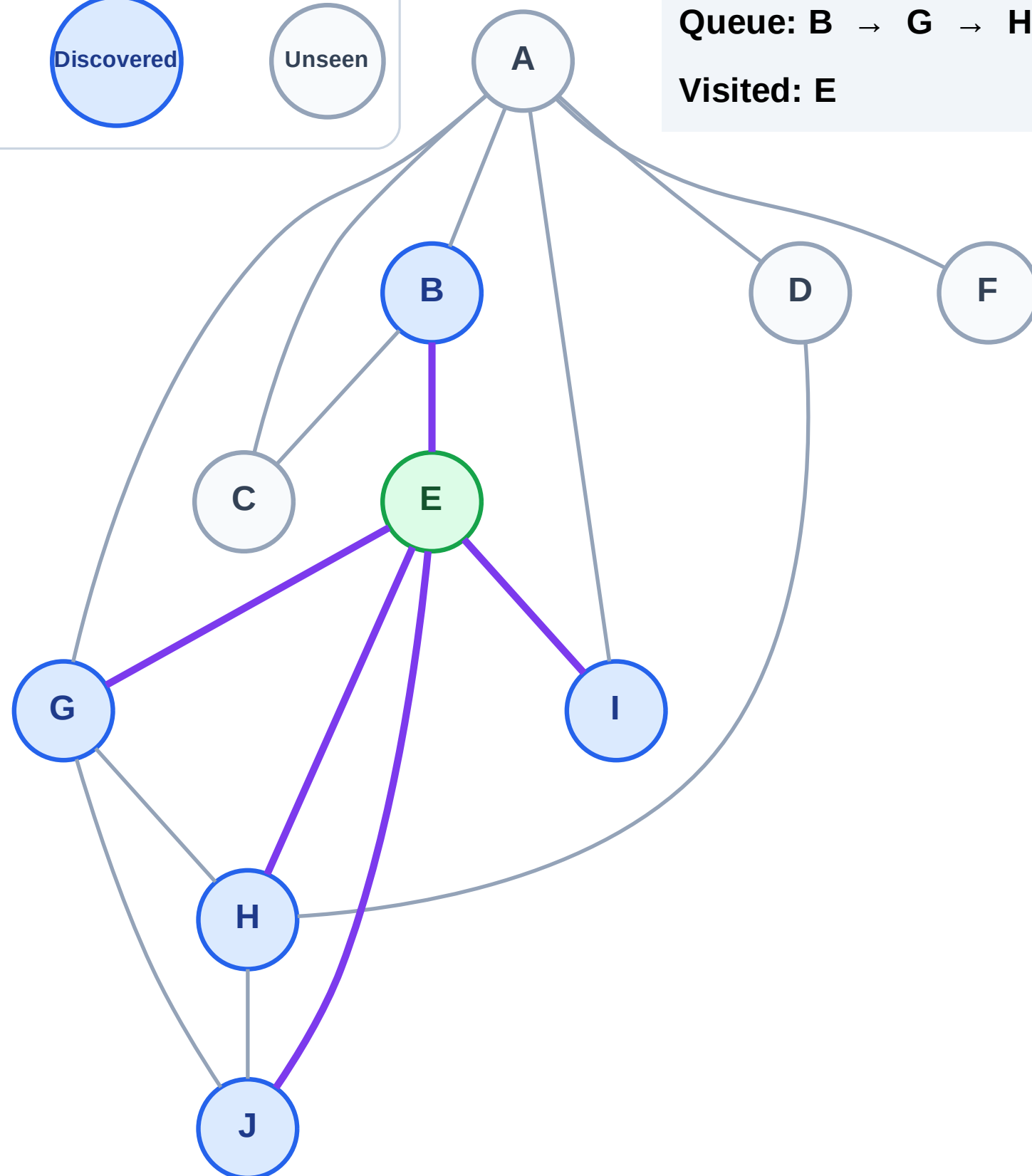
Step 3: Discover B, G, H, I, J from E

Legend



Queue: B → G → H → I → J

Visited: E



BFS Traversal

Step 4: Process node B

Legend

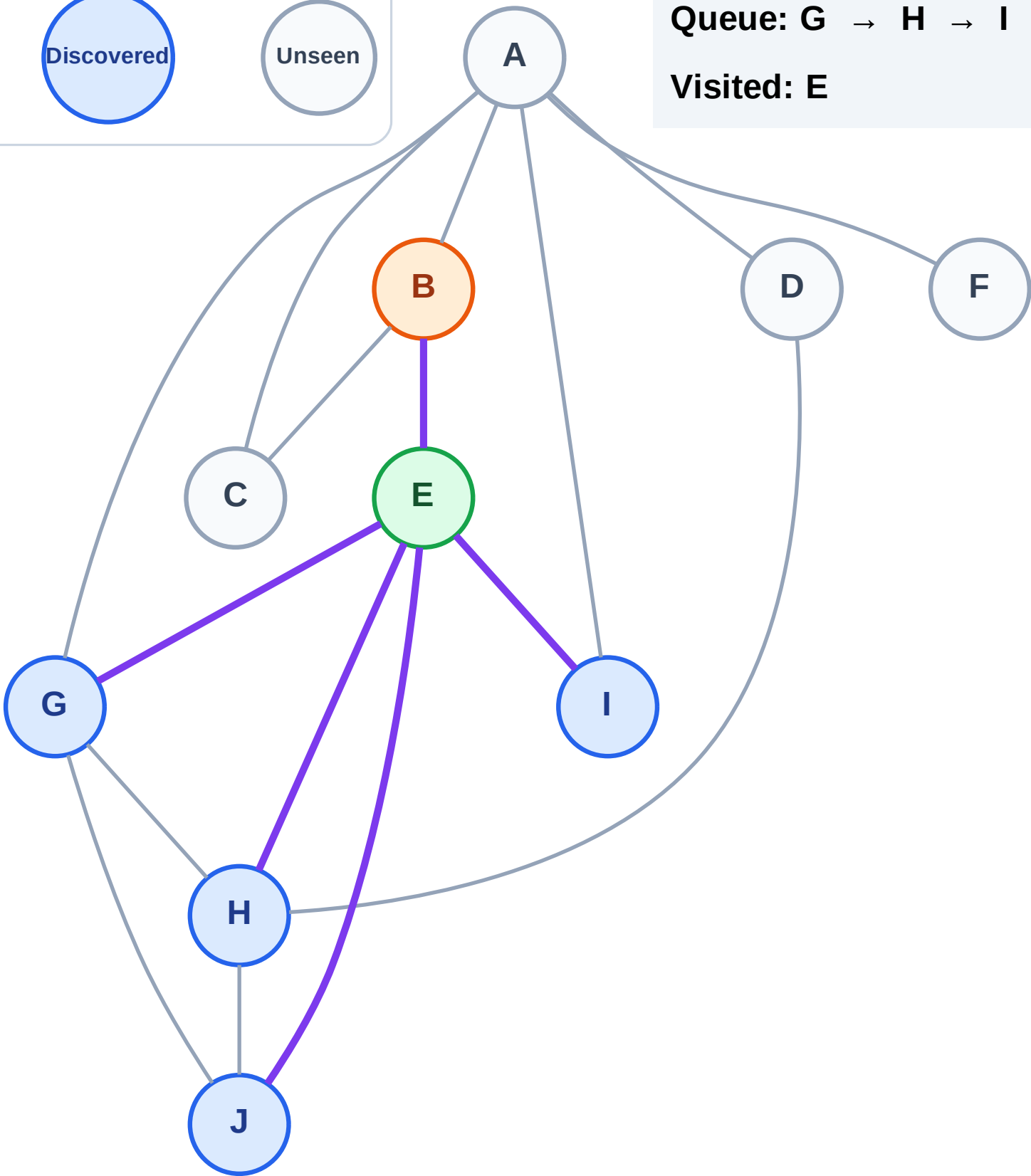
Complete

Processing

Discovered

Unseen

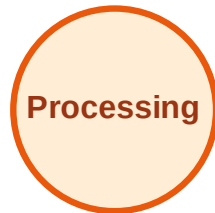
Queue: G → H → I → J
Visited: E



BFS Traversal

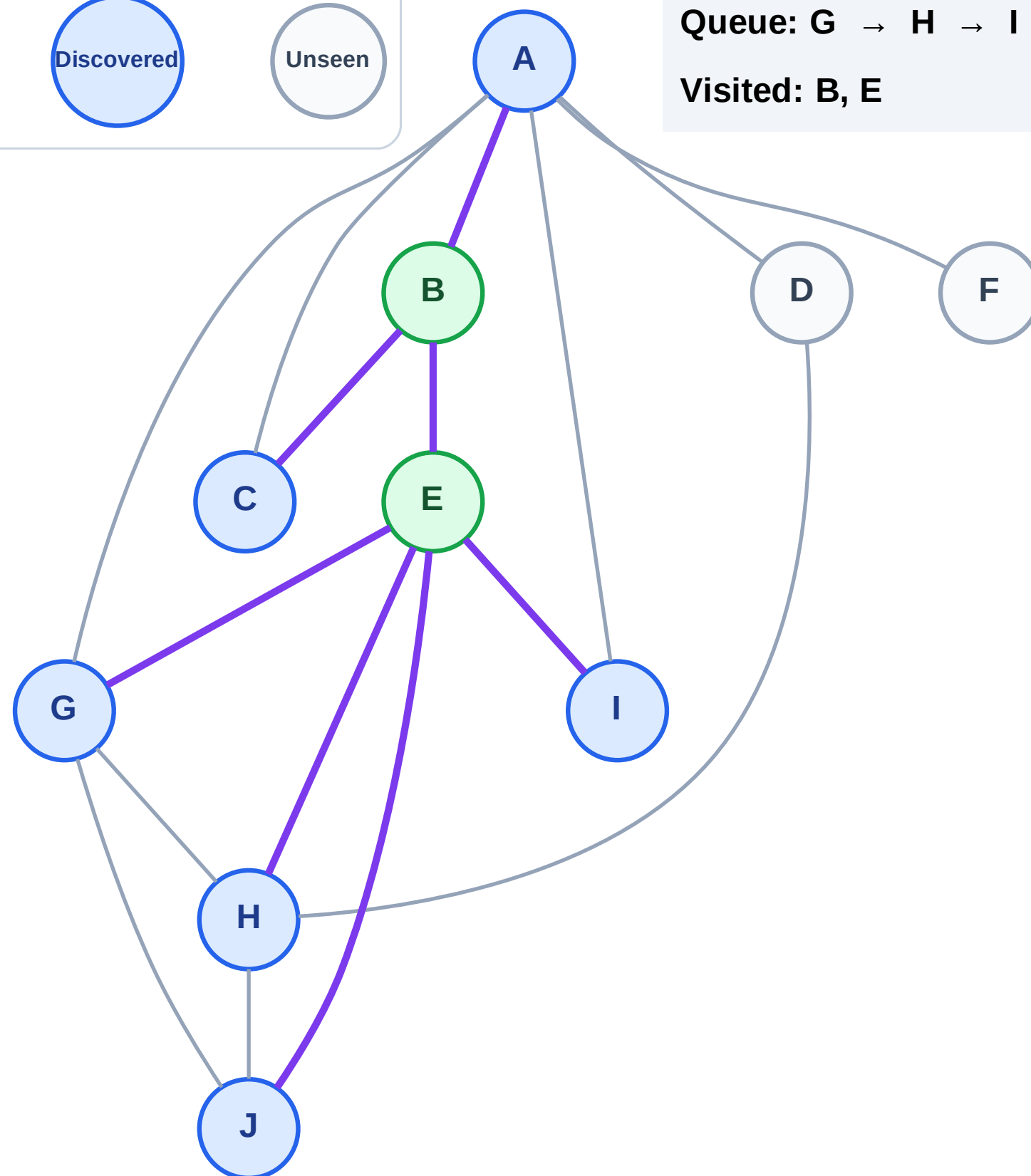
Step 5: Discover A, C from B

Legend



Queue: G → H → I → J → A → C

Visited: B, E



BFS Traversal

Step 6: Process node G

Legend

Complete

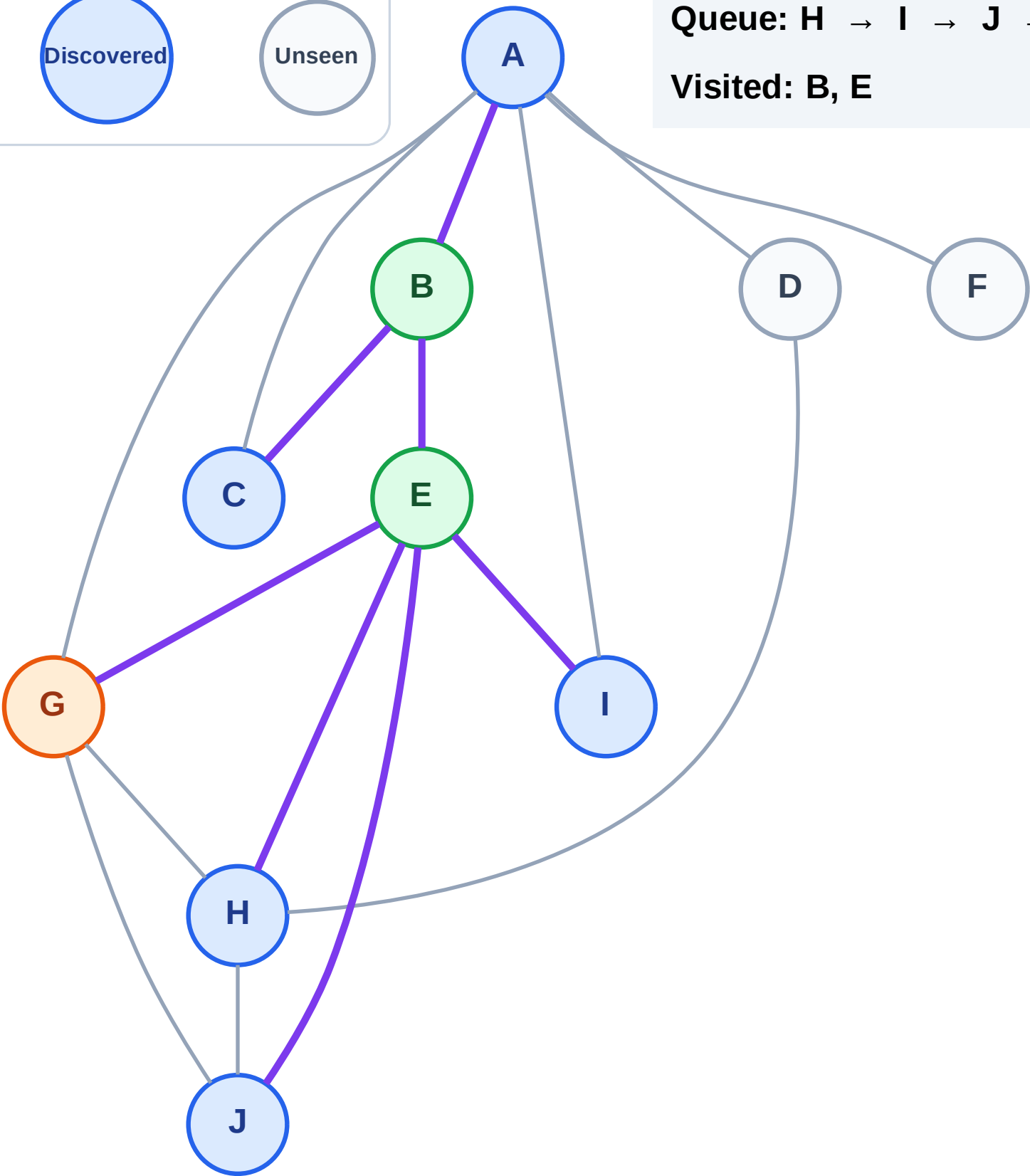
Processing

Discovered

Unseen

Queue: H → I → J → A → C

Visited: B, E



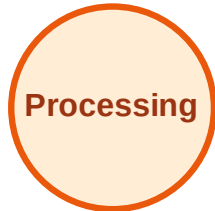
BFS Traversal

Step 7: No new neighbors from G

Legend



Complete



Processing



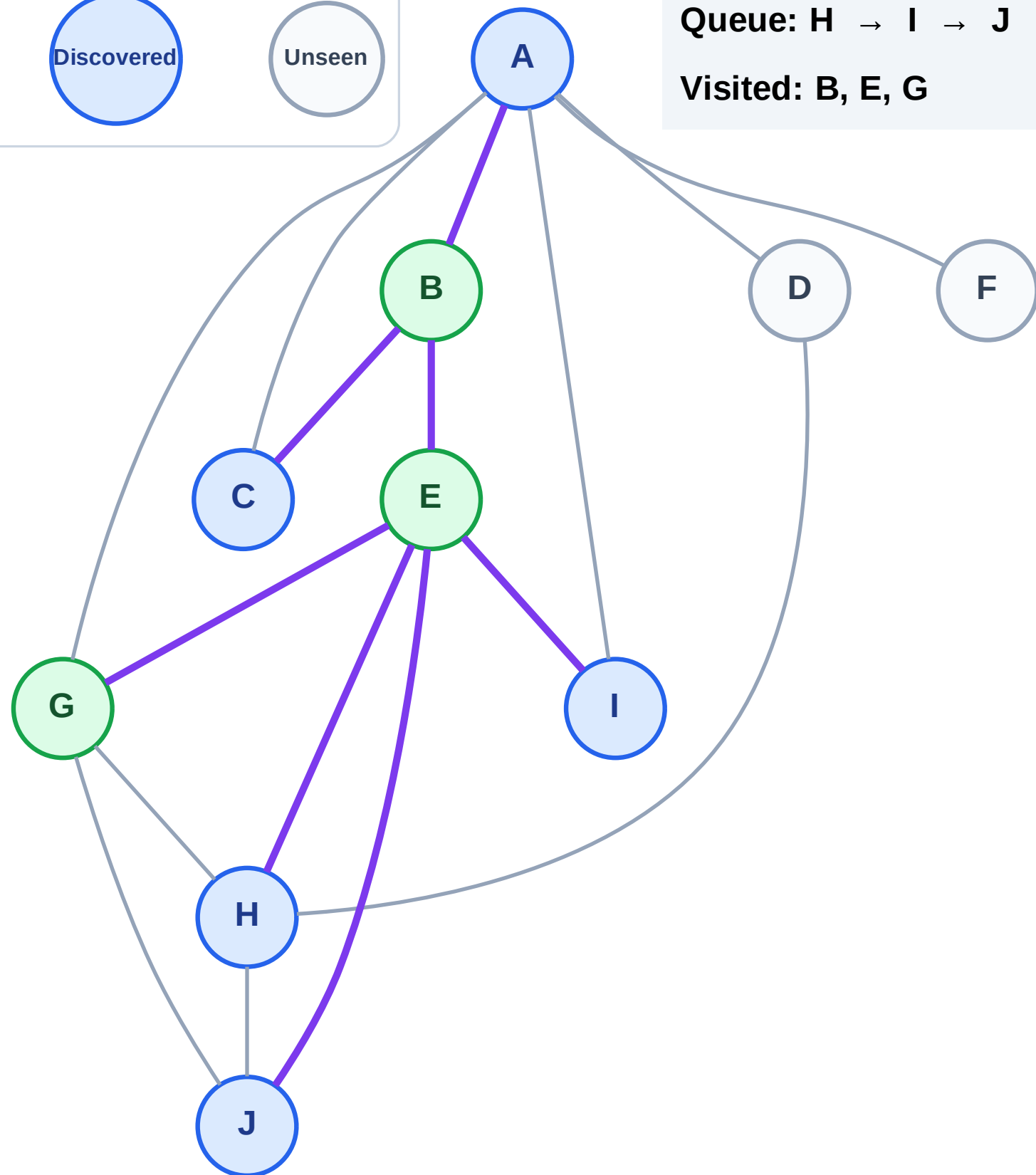
Discovered



Unseen

Queue: H → I → J → A → C

Visited: B, E, G



BFS Traversal

Step 8: Process node H

Legend

Complete

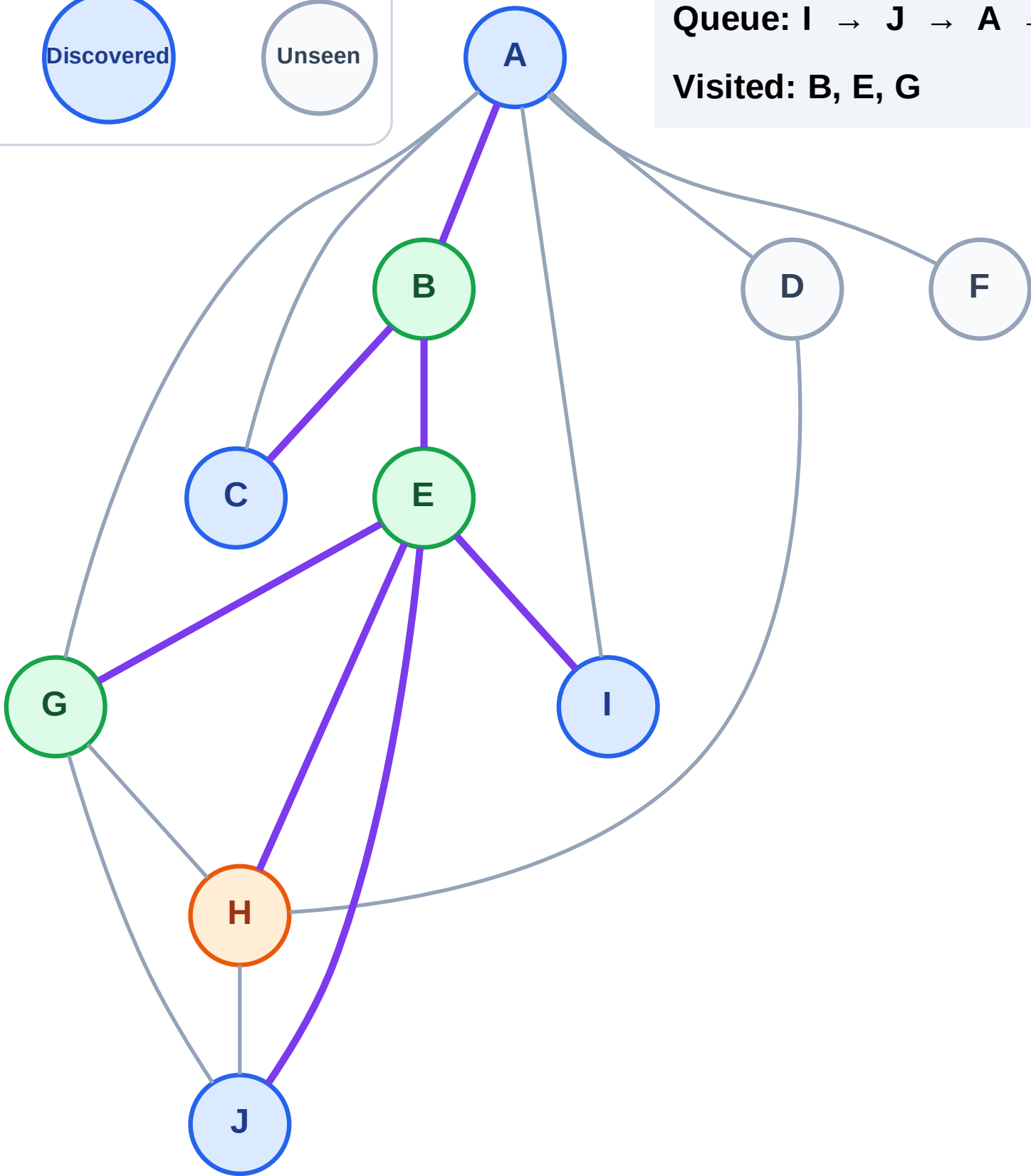
Processing

Discovered

Unseen

Queue: I → J → A → C

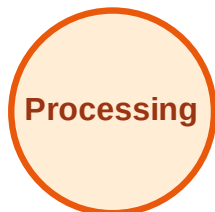
Visited: B, E, G



BFS Traversal

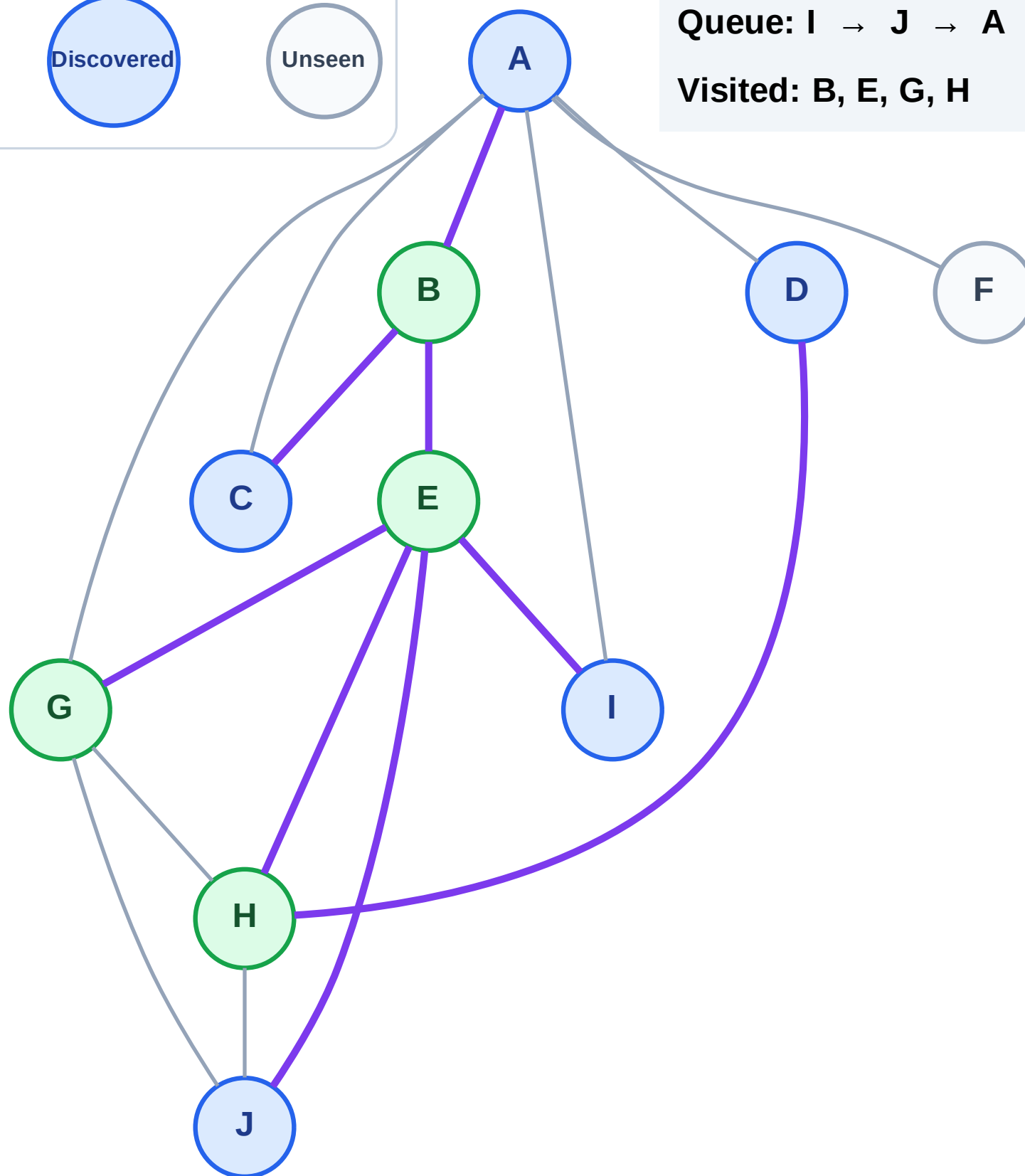
Step 9: Discover D from H

Legend



Queue: I → J → A → C → D

Visited: B, E, G, H



BFS Traversal

Step 10: Process node I

Legend

Complete

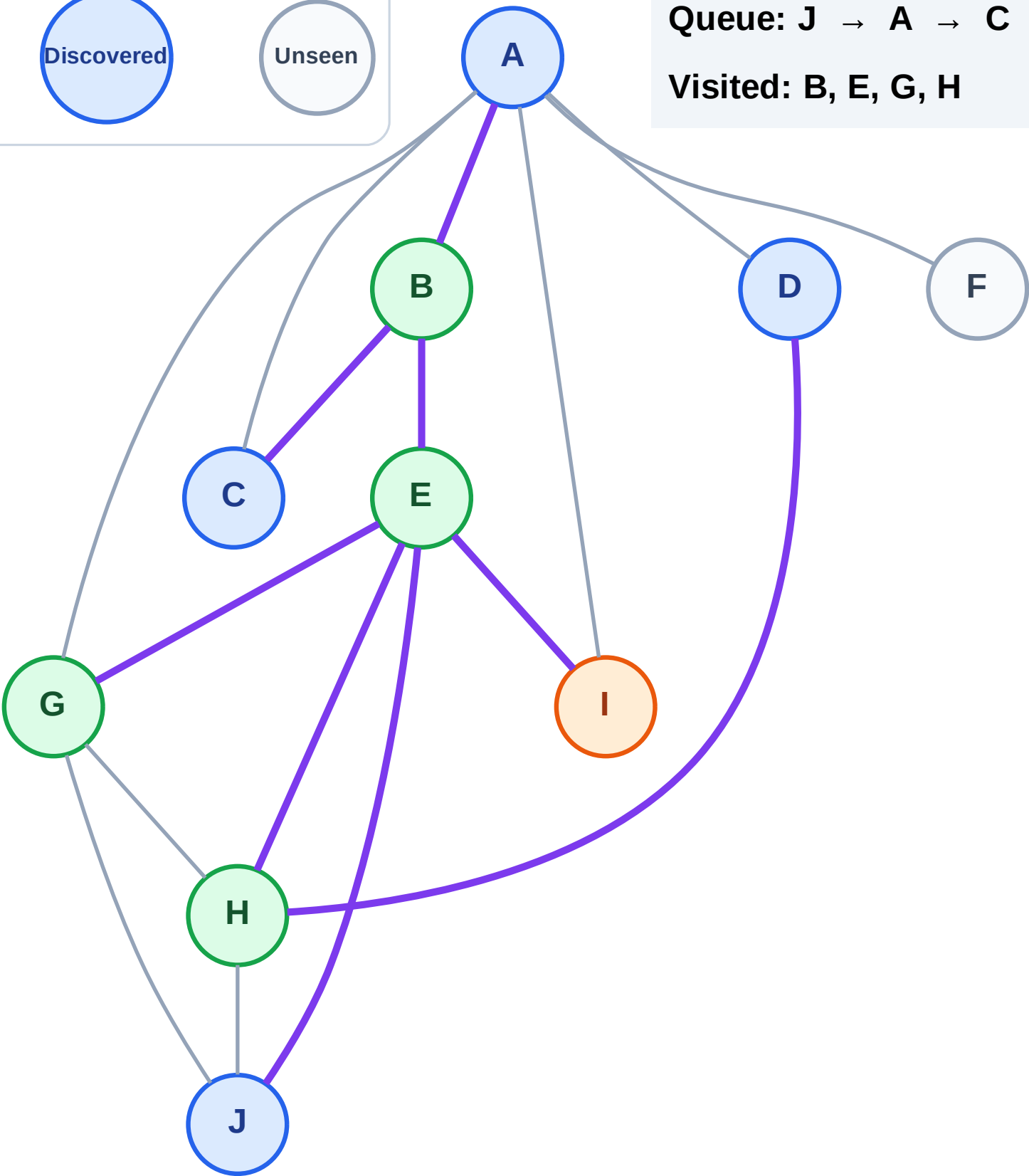
Processing

Discovered

Unseen

Queue: J → A → C → D

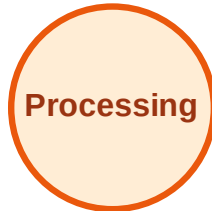
Visited: B, E, G, H



BFS Traversal

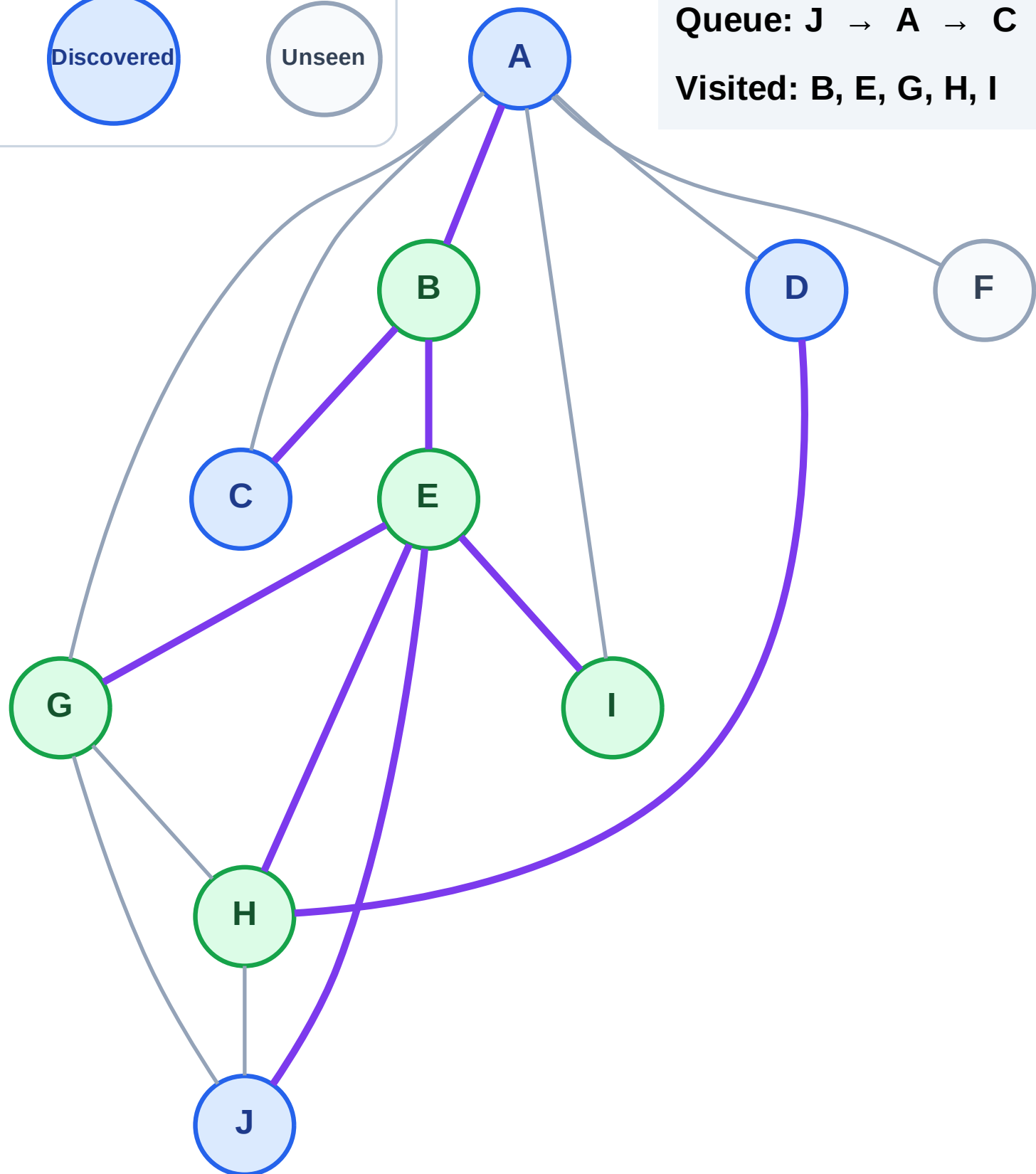
Step 11: No new neighbors from I

Legend



Queue: J → A → C → D

Visited: B, E, G, H, I



BFS Traversal

Step 12: Process node J

Legend

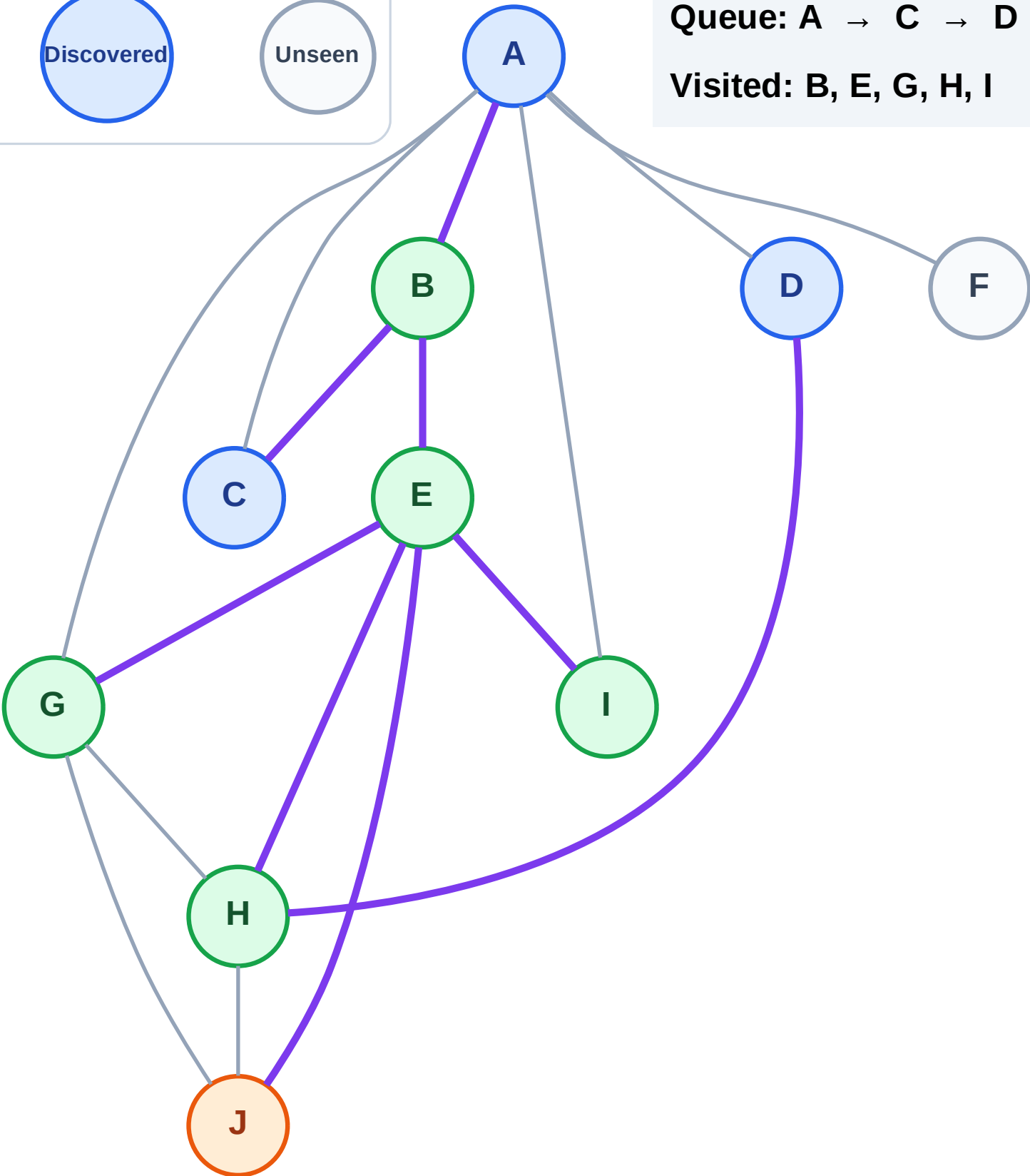
Complete

Processing

Discovered

Unseen

Queue: A → C → D
Visited: B, E, G, H, I



BFS Traversal

Step 13: No new neighbors from J

Legend

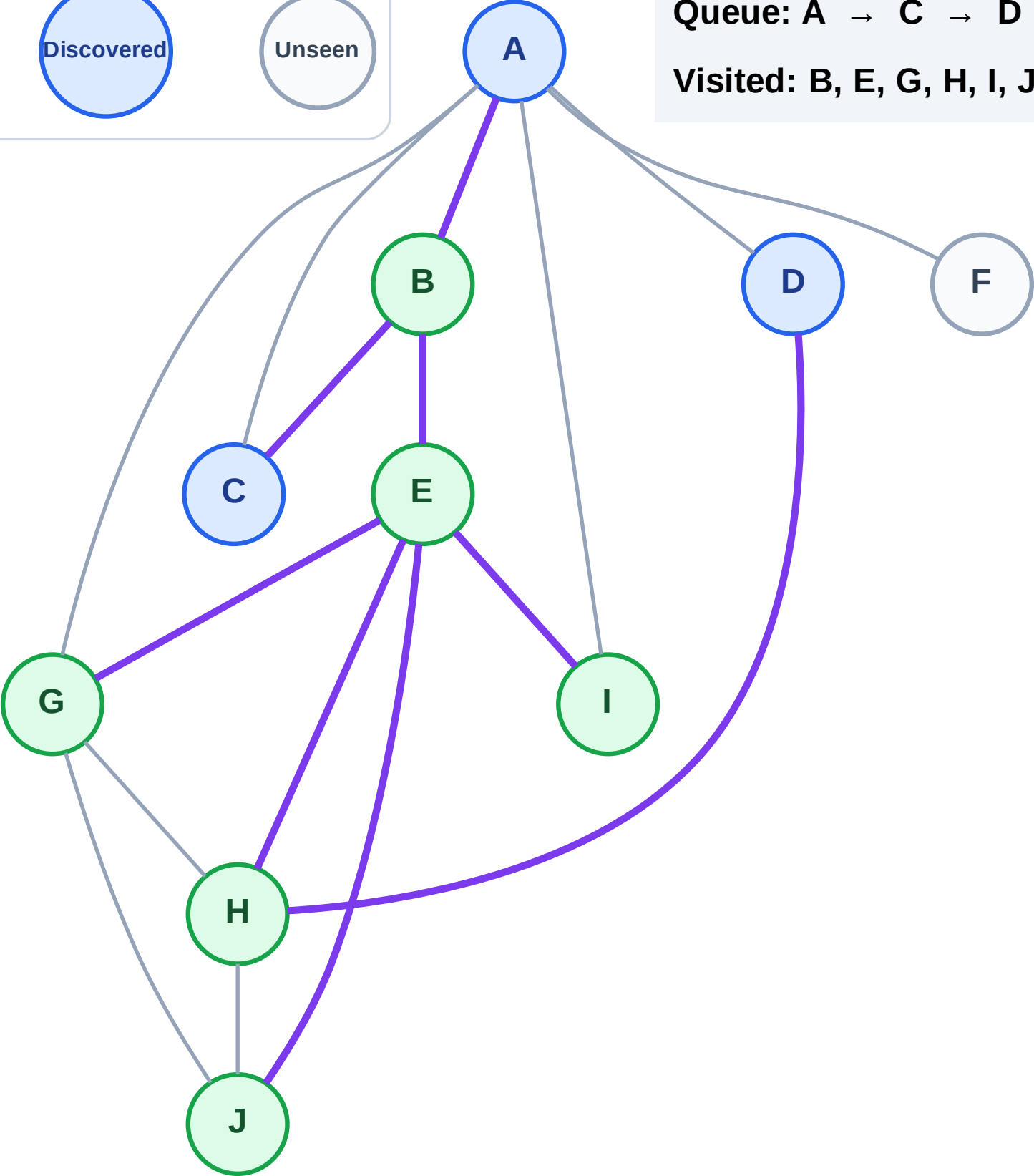
Complete

Processing

Discovered

Unseen

Queue: A → C → D
Visited: B, E, G, H, I, J



BFS Traversal

Step 14: Process node A

Legend

Complete

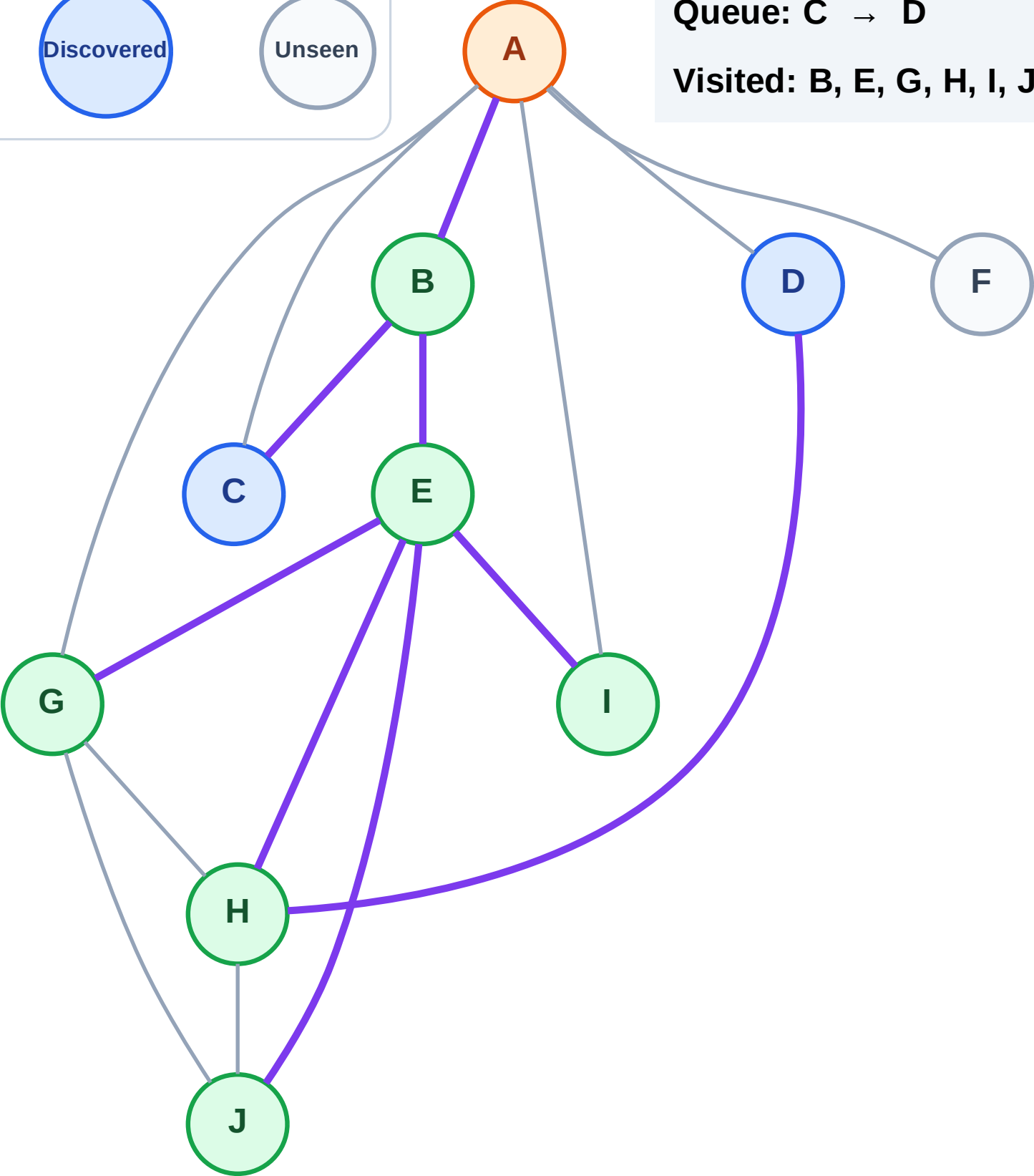
Processing

Discovered

Unseen

Queue: C → D

Visited: B, E, G, H, I, J



BFS Traversal

Step 15: Discover F from A

Legend

Complete

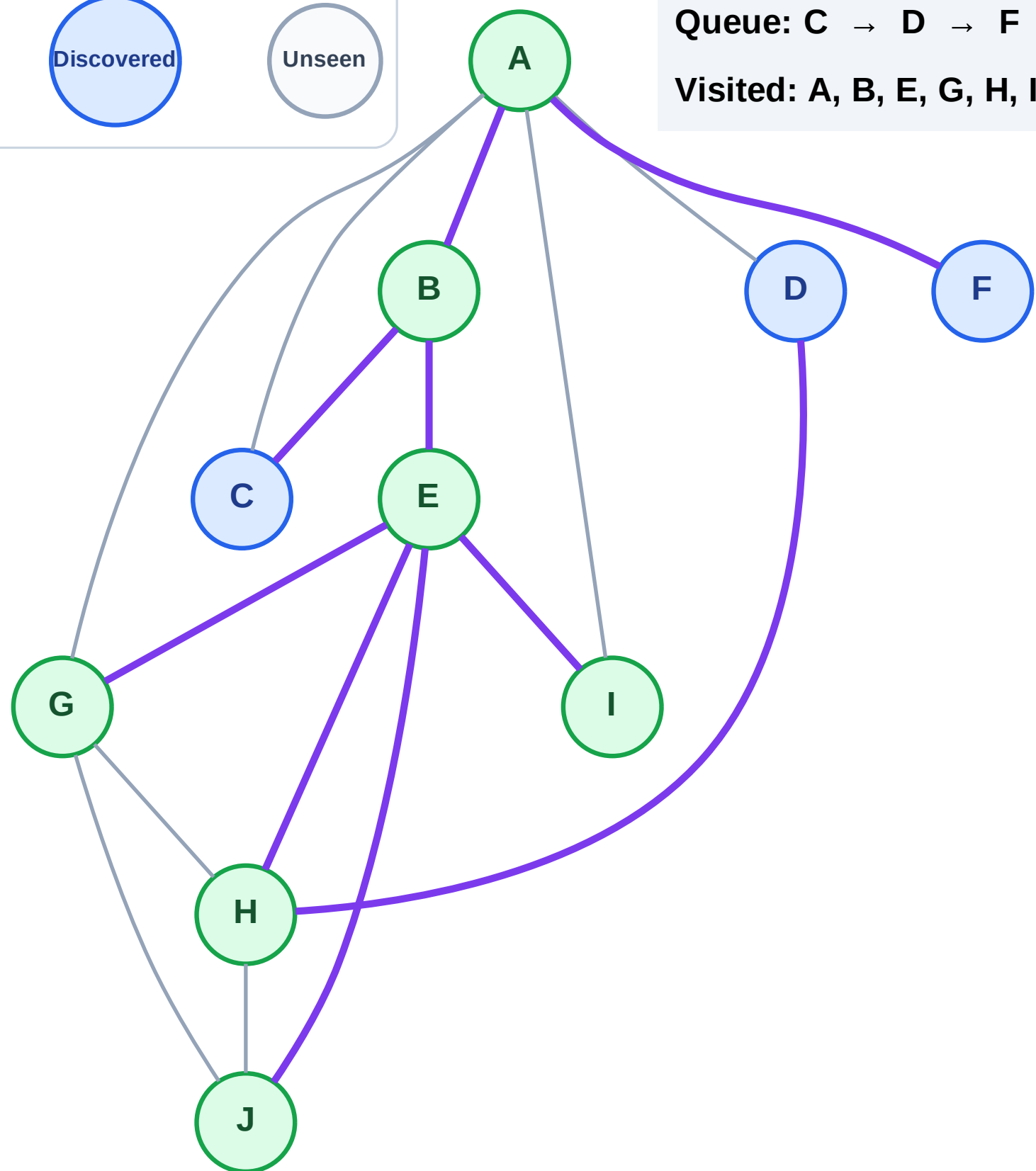
Processing

Discovered

Unseen

Queue: C → D → F

Visited: A, B, E, G, H, I, J



BFS Traversal

Step 16: Process node C

Legend

Complete

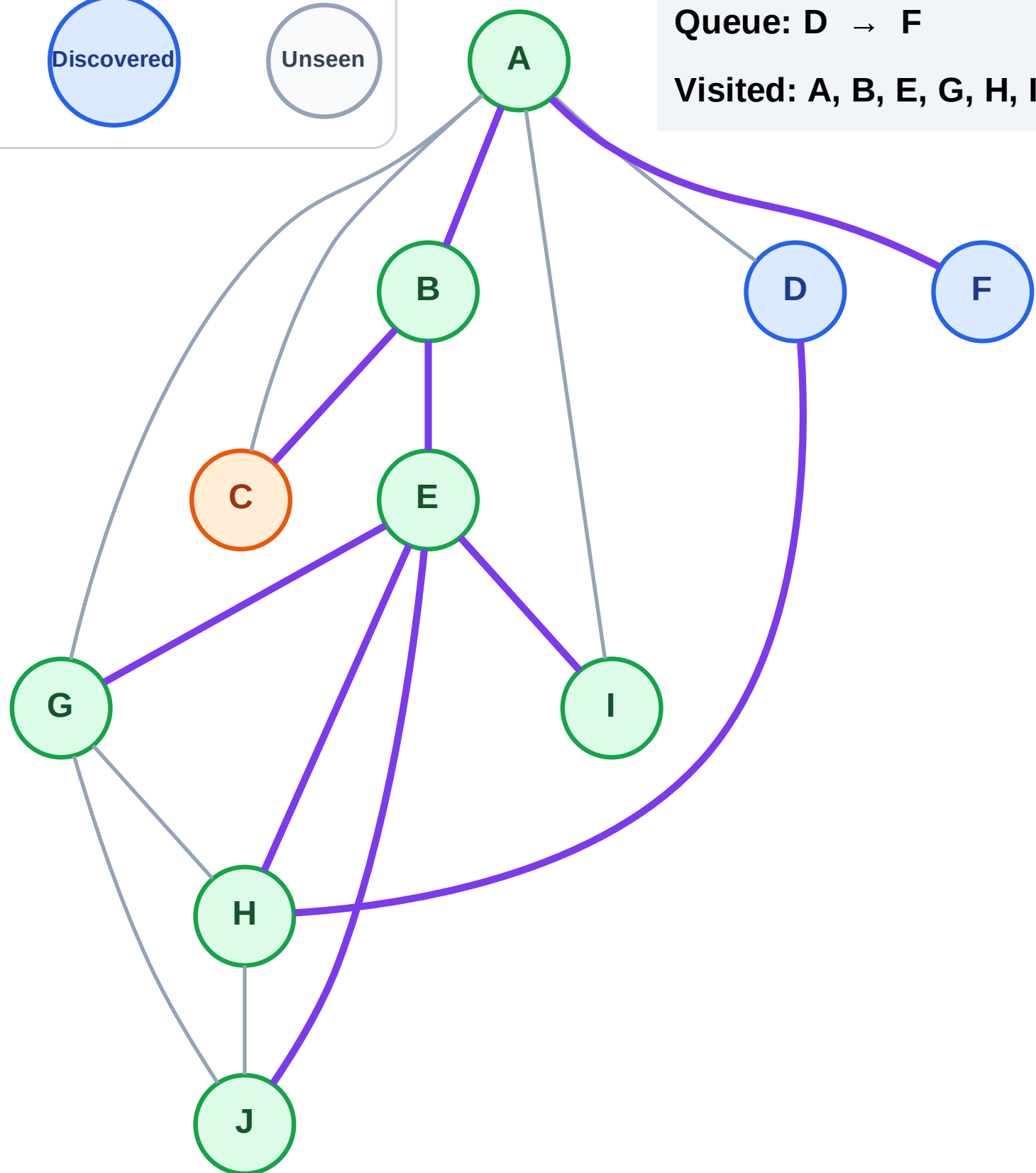
Processing

Discovered

Unseen

Queue: D → F

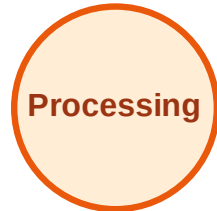
Visited: A, B, E, G, H, I, J



BFS Traversal

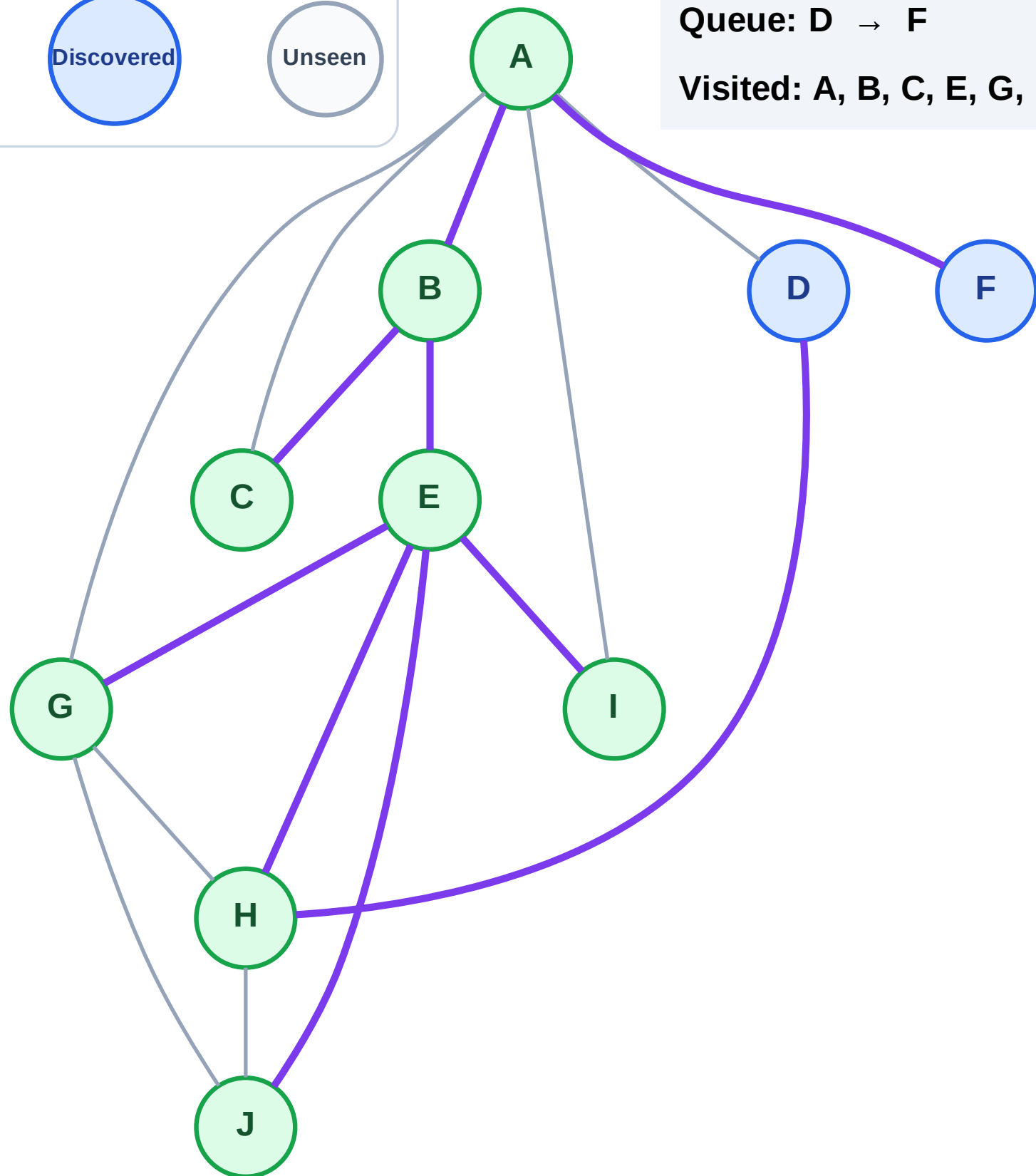
Step 17: No new neighbors from C

Legend



Queue: D → F

Visited: A, B, C, E, G, H, I, J



BFS Traversal

Step 18: Process node D

Legend

Complete

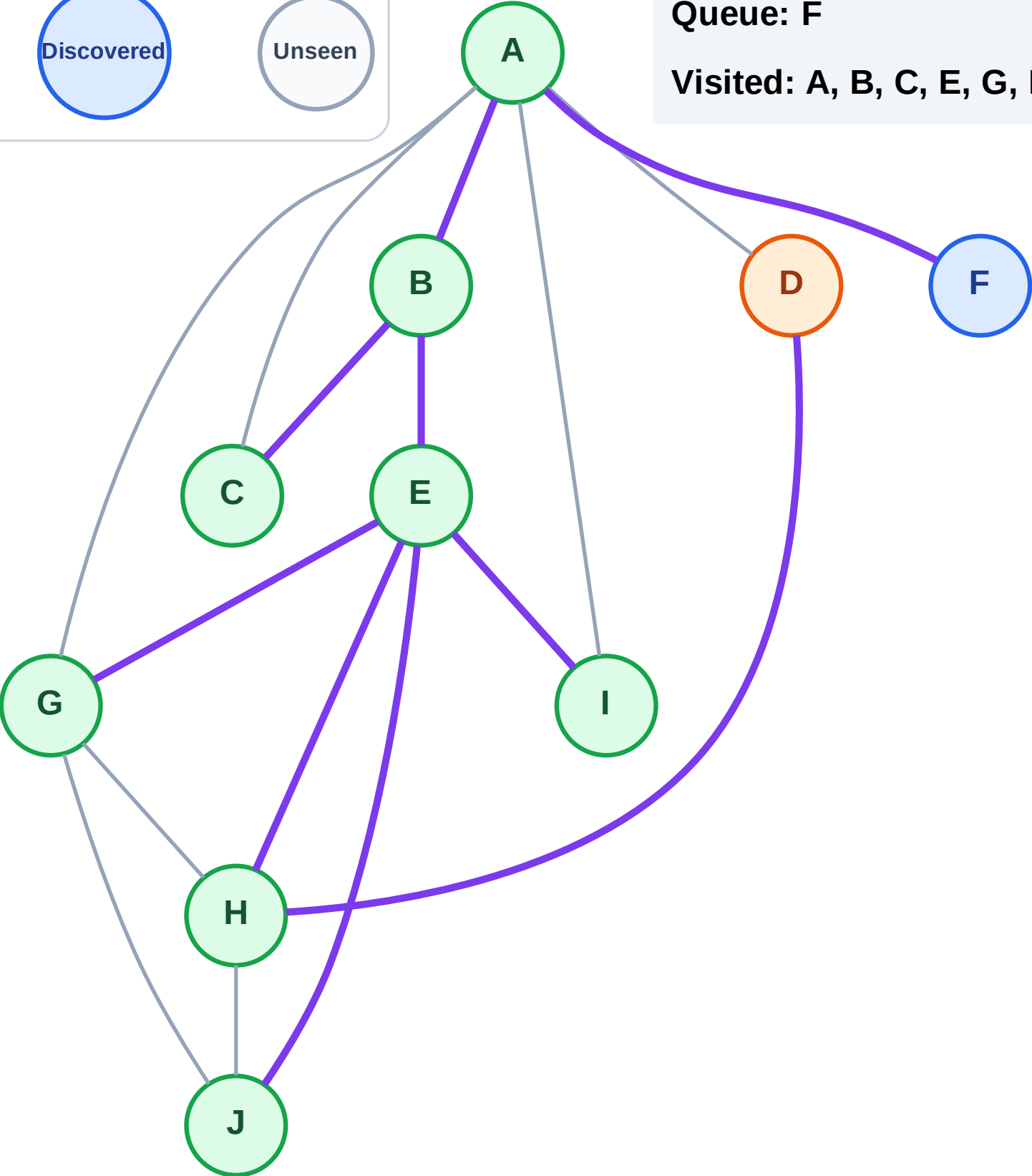
Processing

Discovered

Unseen

Queue: F

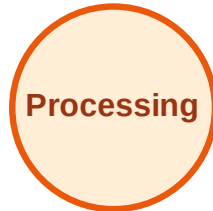
Visited: A, B, C, E, G, H, I, J



BFS Traversal

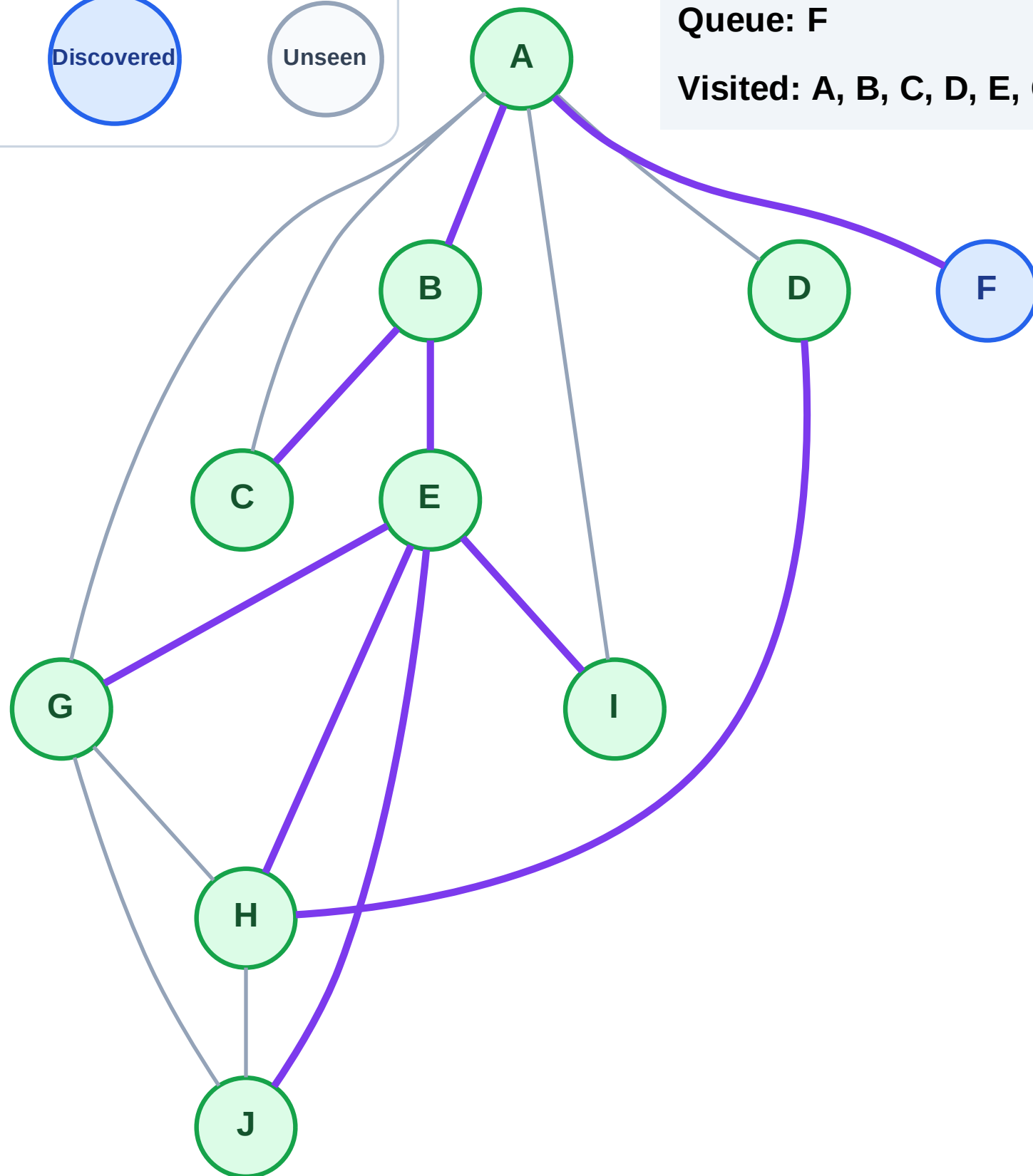
Step 19: No new neighbors from D

Legend



Queue: F

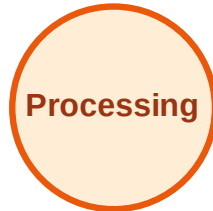
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

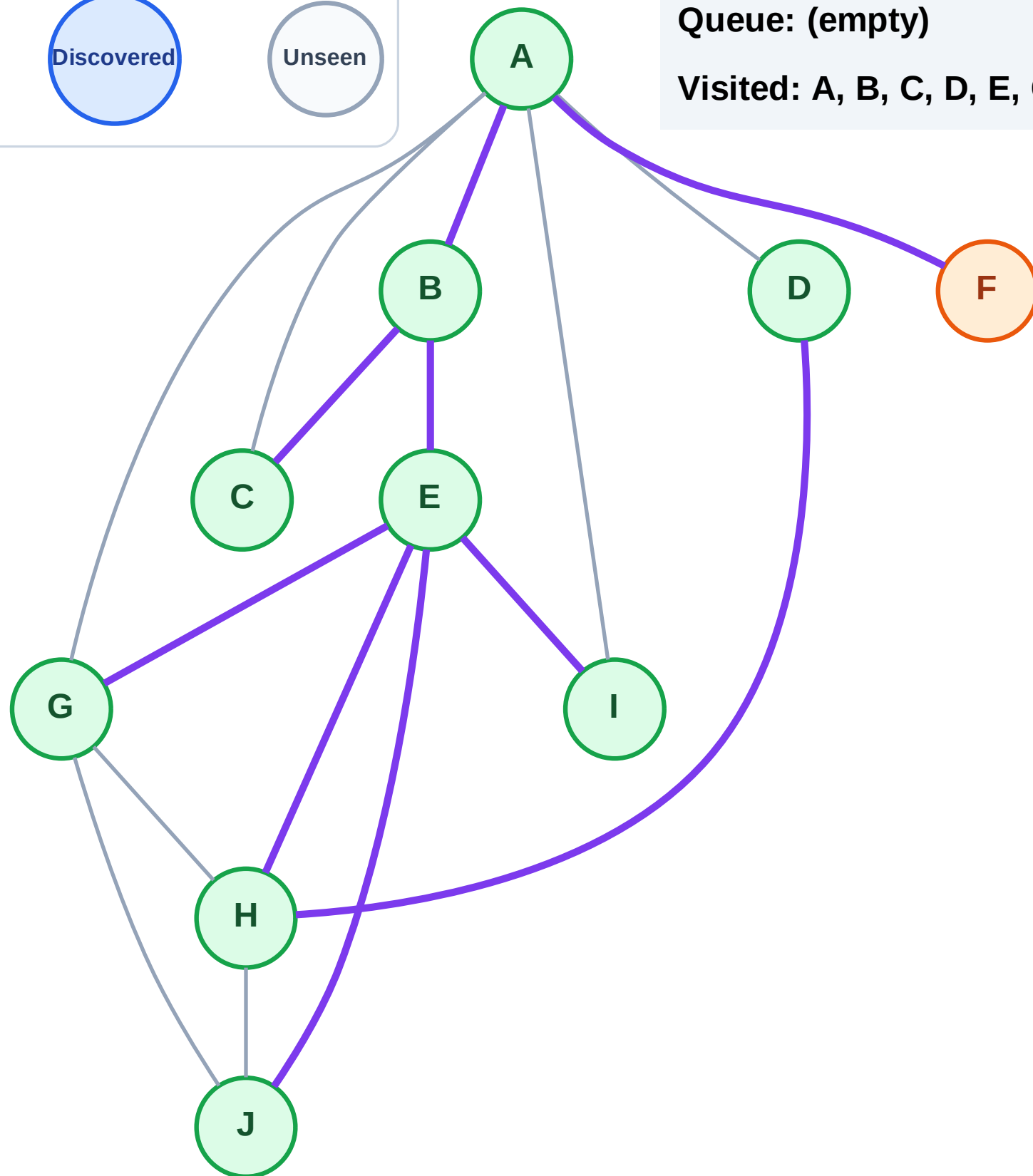
Step 20: Process node F

Legend



Queue: (empty)

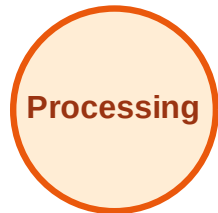
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

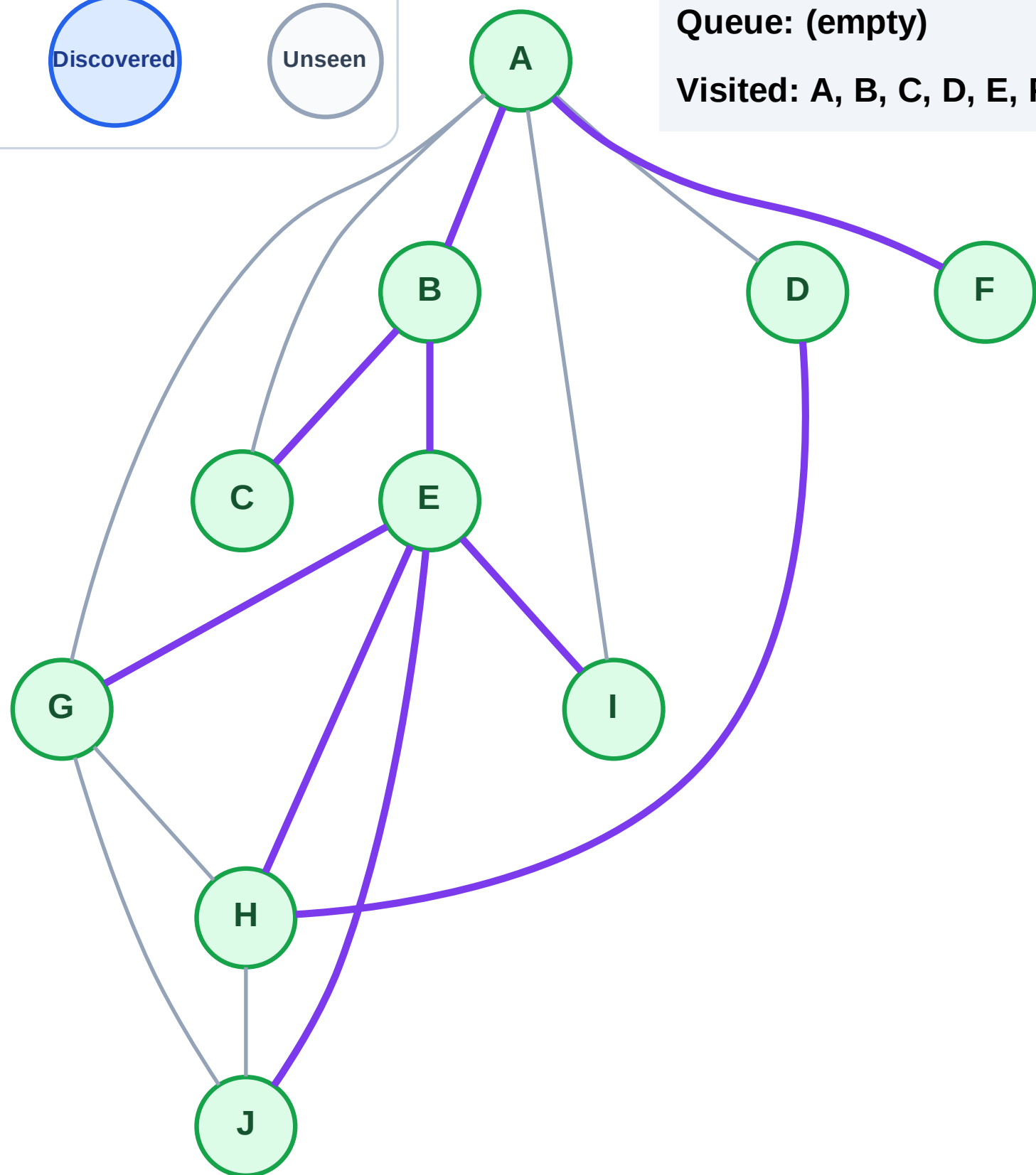
Step 21: No new neighbors from F

Legend



Queue: (empty)

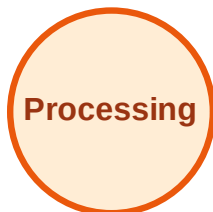
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

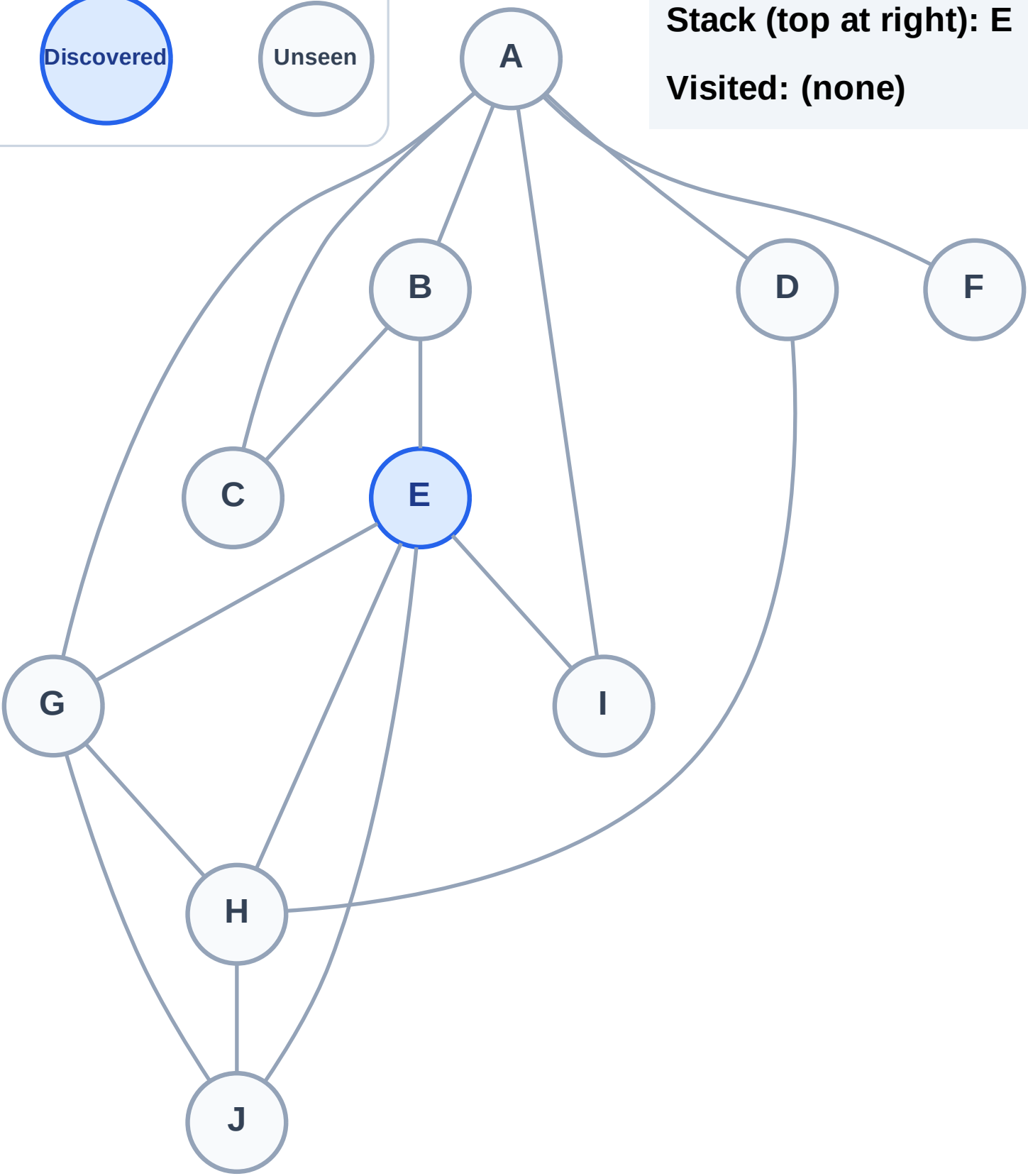
Step 1: Start at E

Legend



Stack (top at right): E

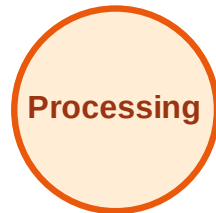
Visited: (none)



DFS Traversal

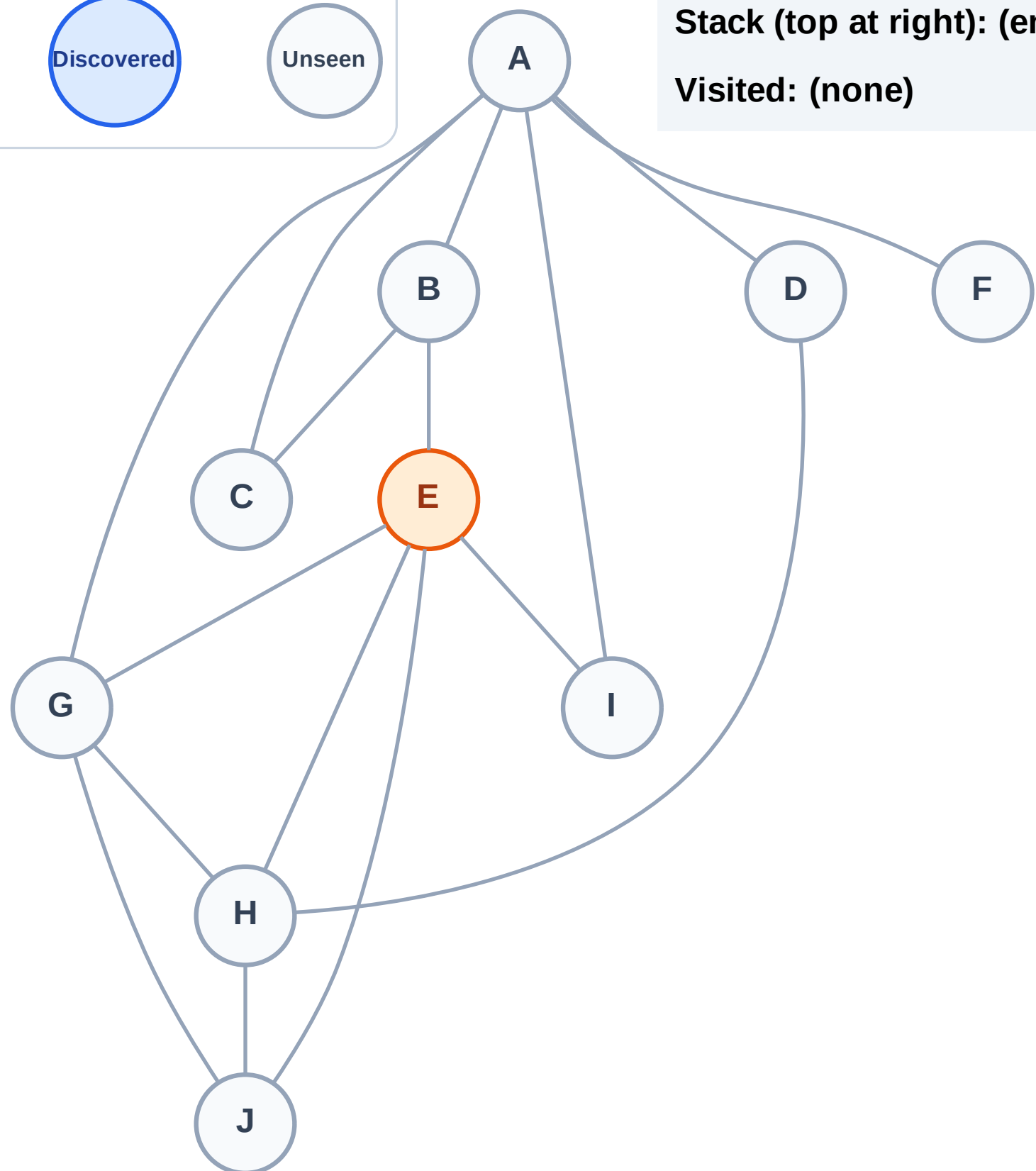
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



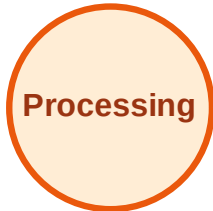
DFS Traversal

Step 3: Discover J, I, H, G, B from E

Legend



Complete



Processing



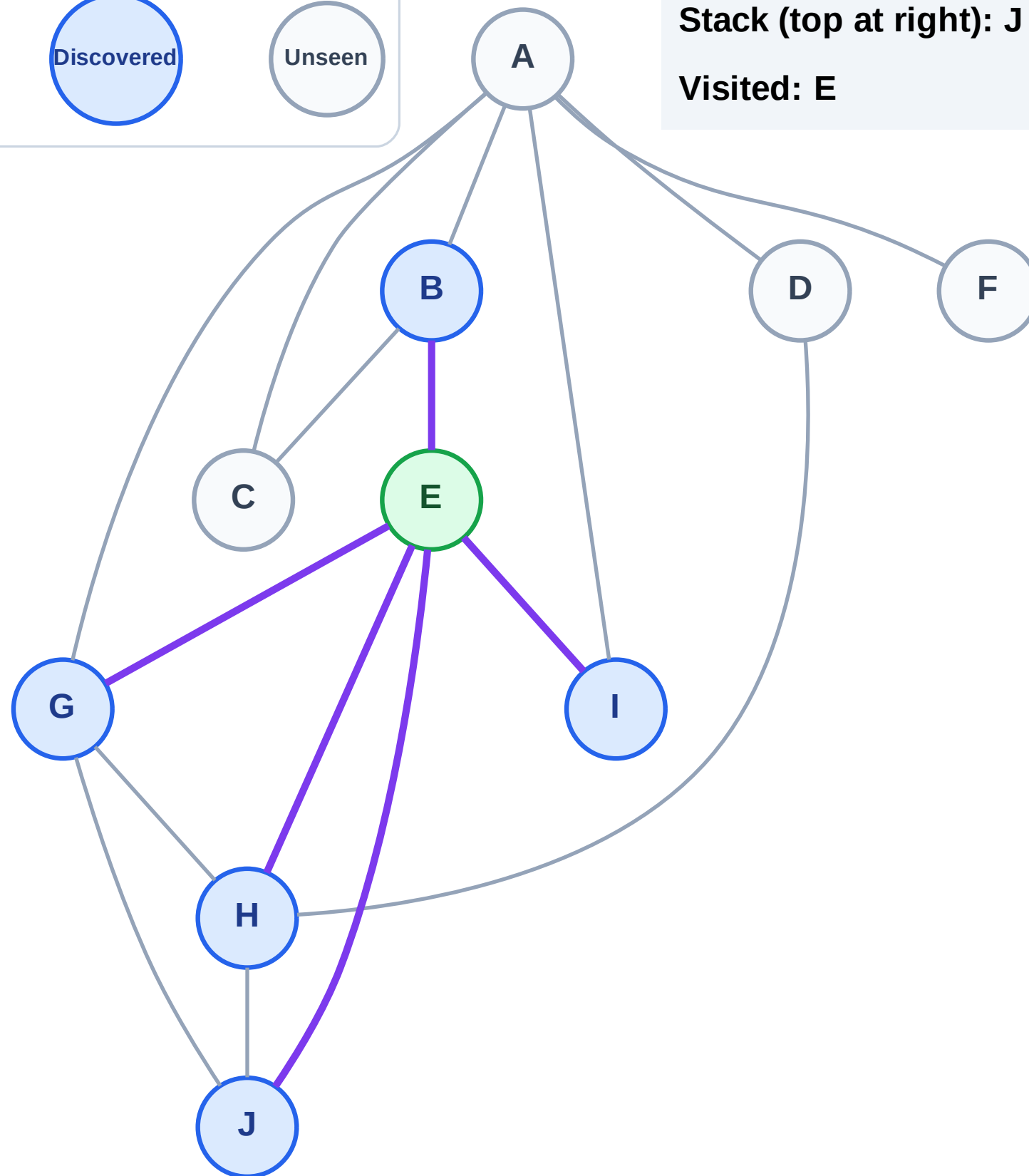
Discovered



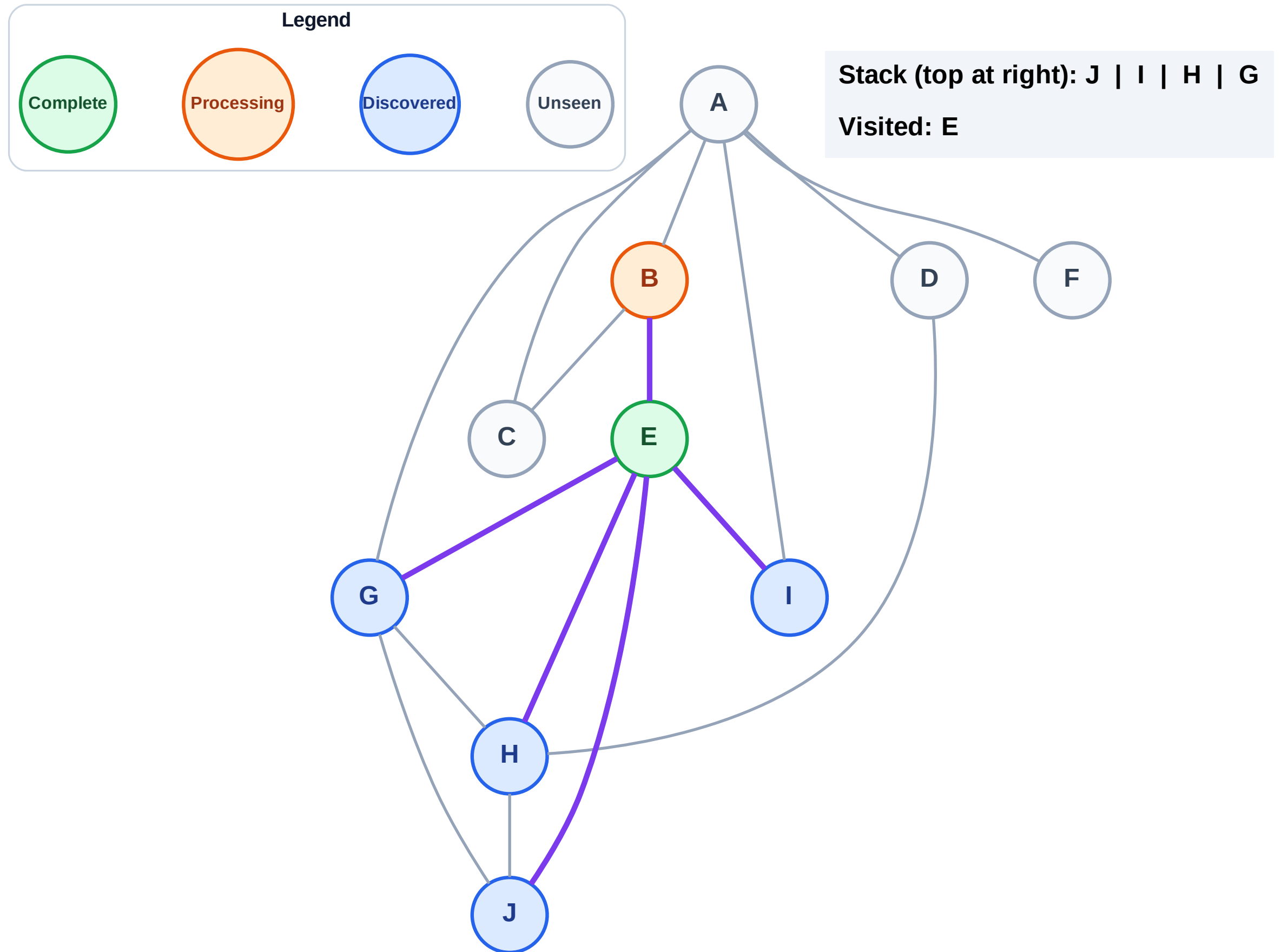
Unseen

Stack (top at right): J | I | H | G | B

Visited: E



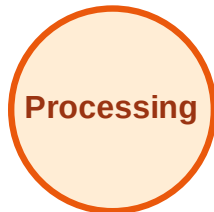
Step 4: Process node B



DFS Traversal

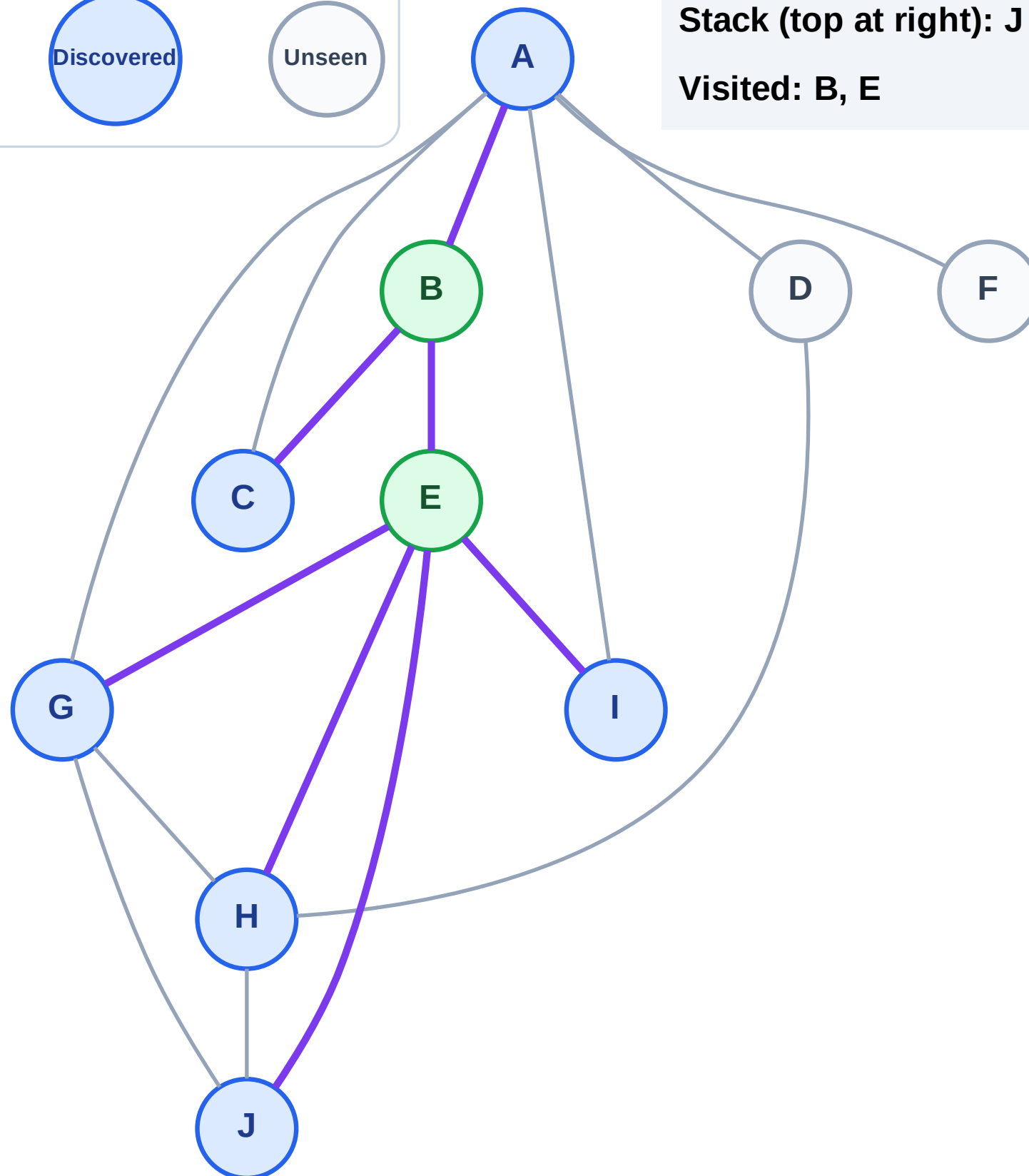
Step 5: Discover C, A from B

Legend



Stack (top at right): J | I | H | G | C | A

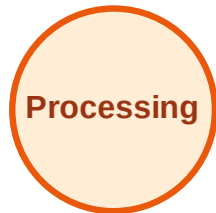
Visited: B, E



DFS Traversal

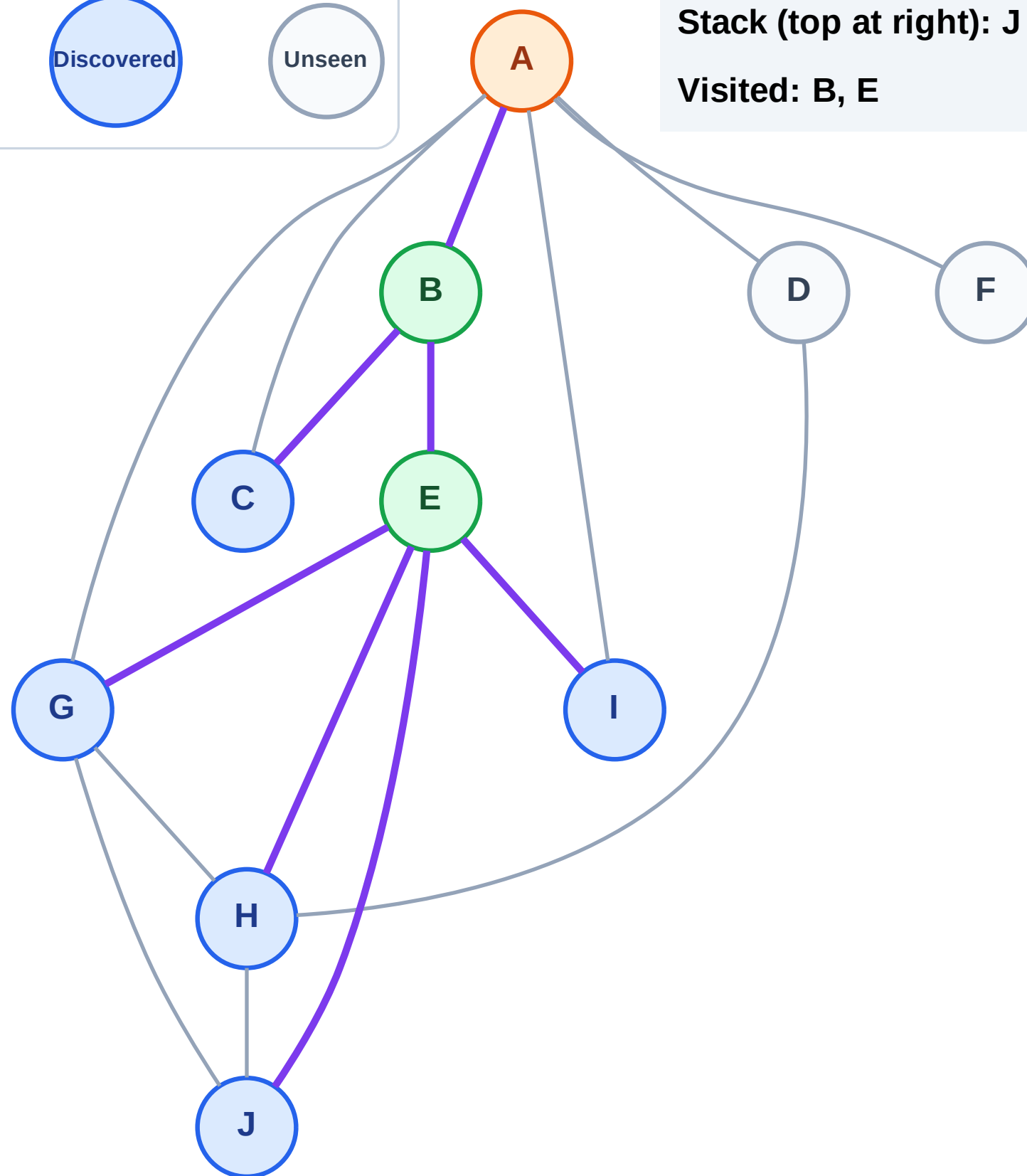
Step 6: Process node A

Legend



Stack (top at right): J | I | H | G | C

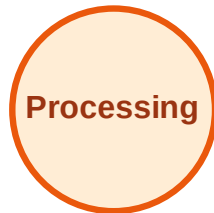
Visited: B, E



DFS Traversal

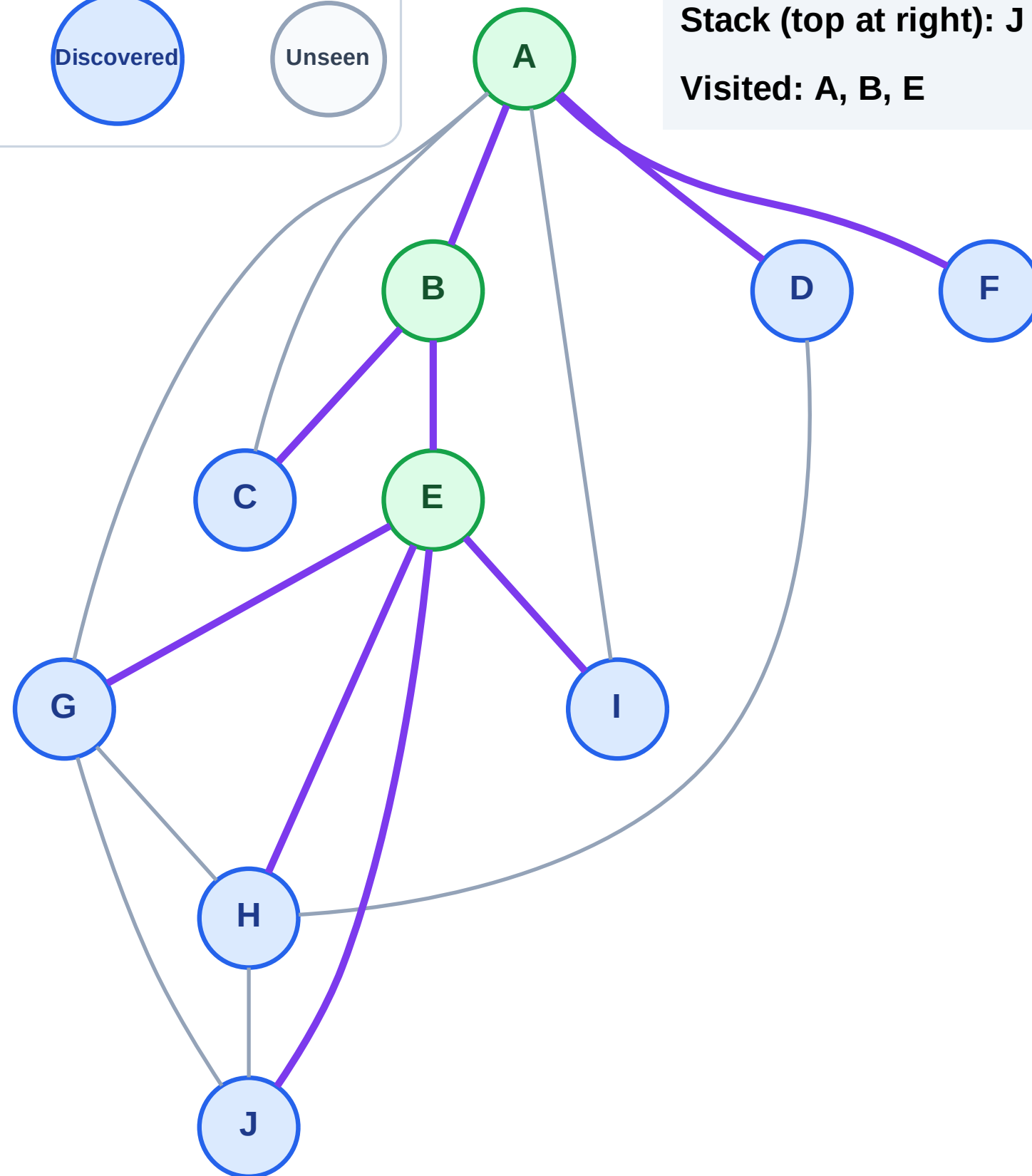
Step 7: Discover F, D from A

Legend



Stack (top at right): J | I | H | G | C | F | D

Visited: A, B, E



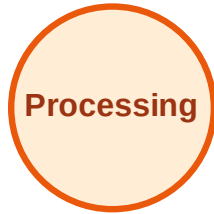
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



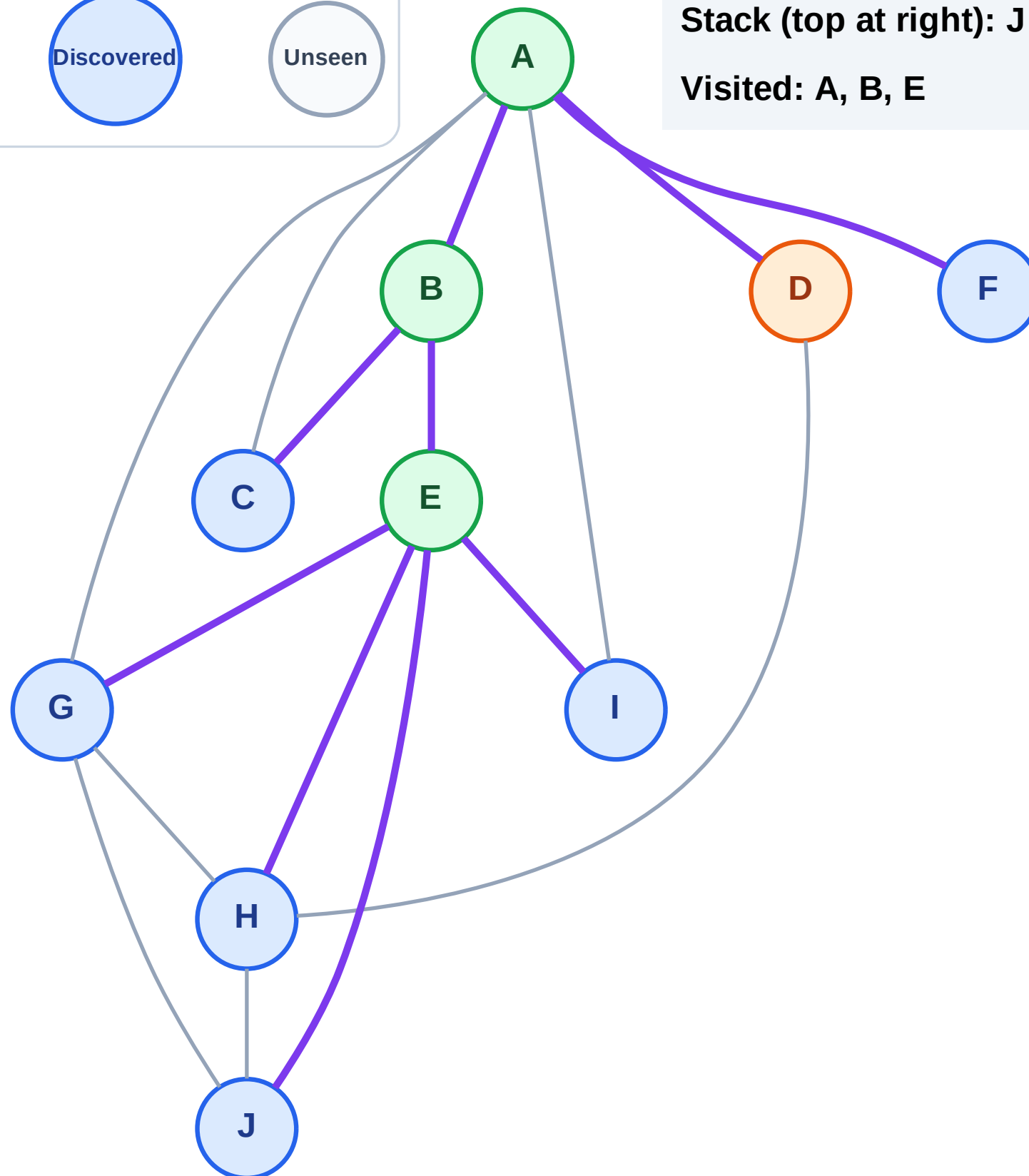
Discovered



Unseen

Stack (top at right): J | I | H | G | C | F

Visited: A, B, E



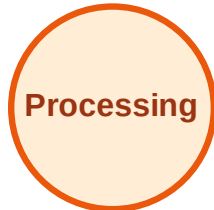
DFS Traversal

Step 9: No new neighbors from D

Legend



Complete



Processing



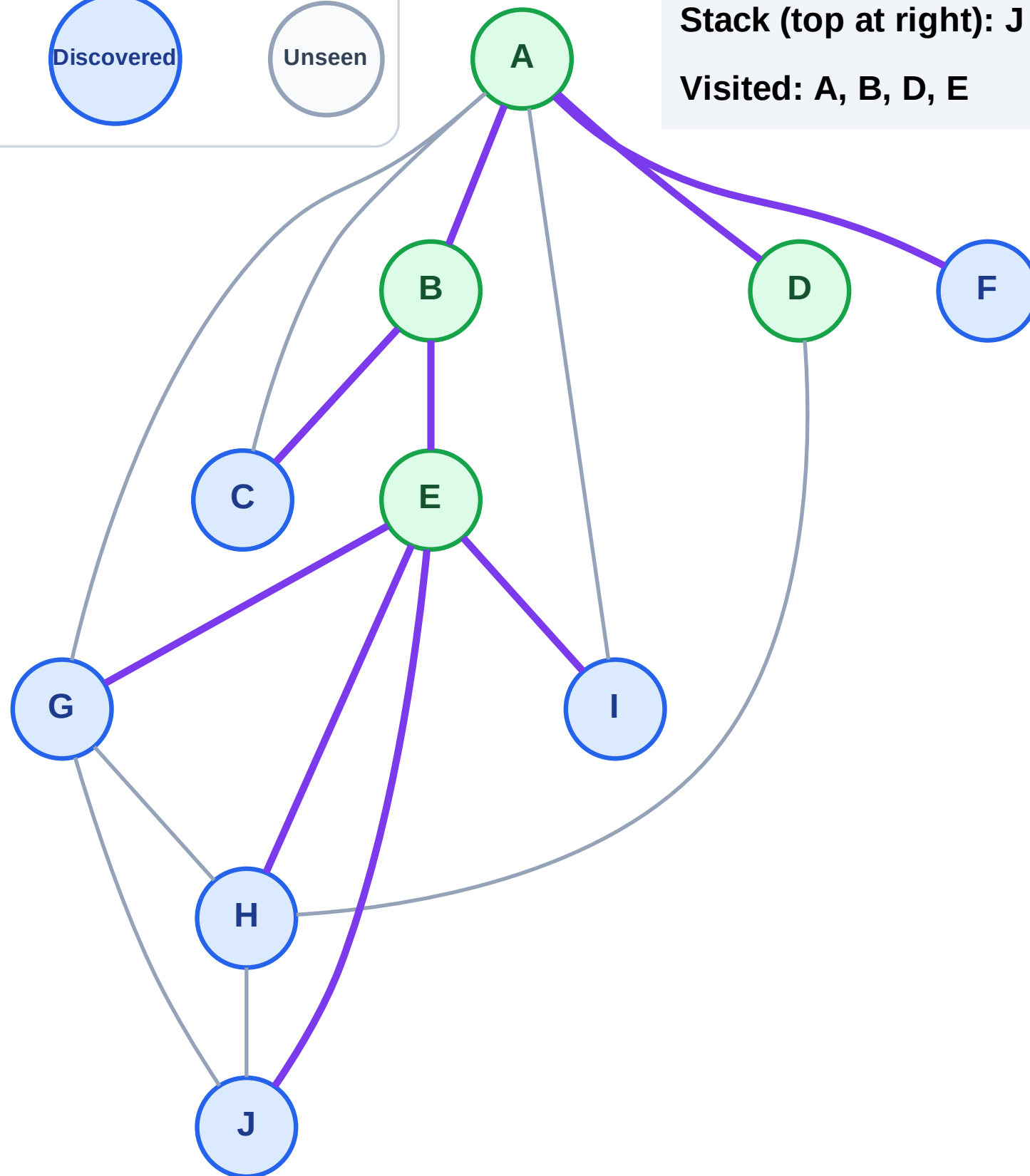
Discovered



Unseen

Stack (top at right): J | I | H | G | C | F

Visited: A, B, D, E



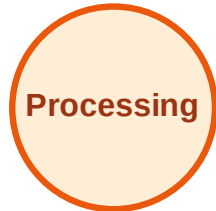
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



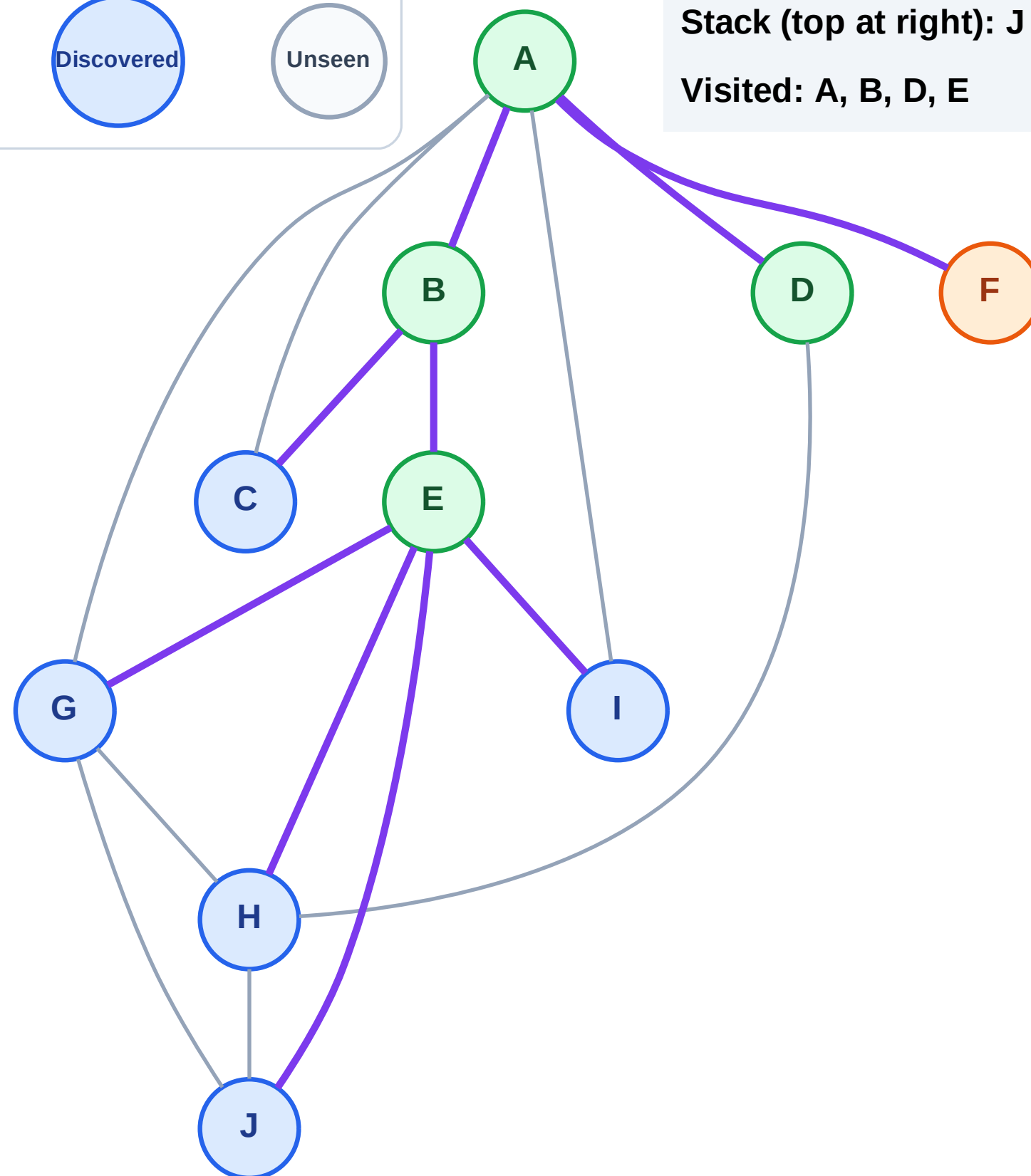
Discovered



Unseen

Stack (top at right): J | I | H | G | C

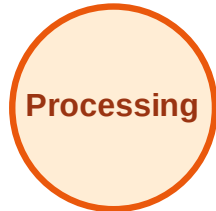
Visited: A, B, D, E



DFS Traversal

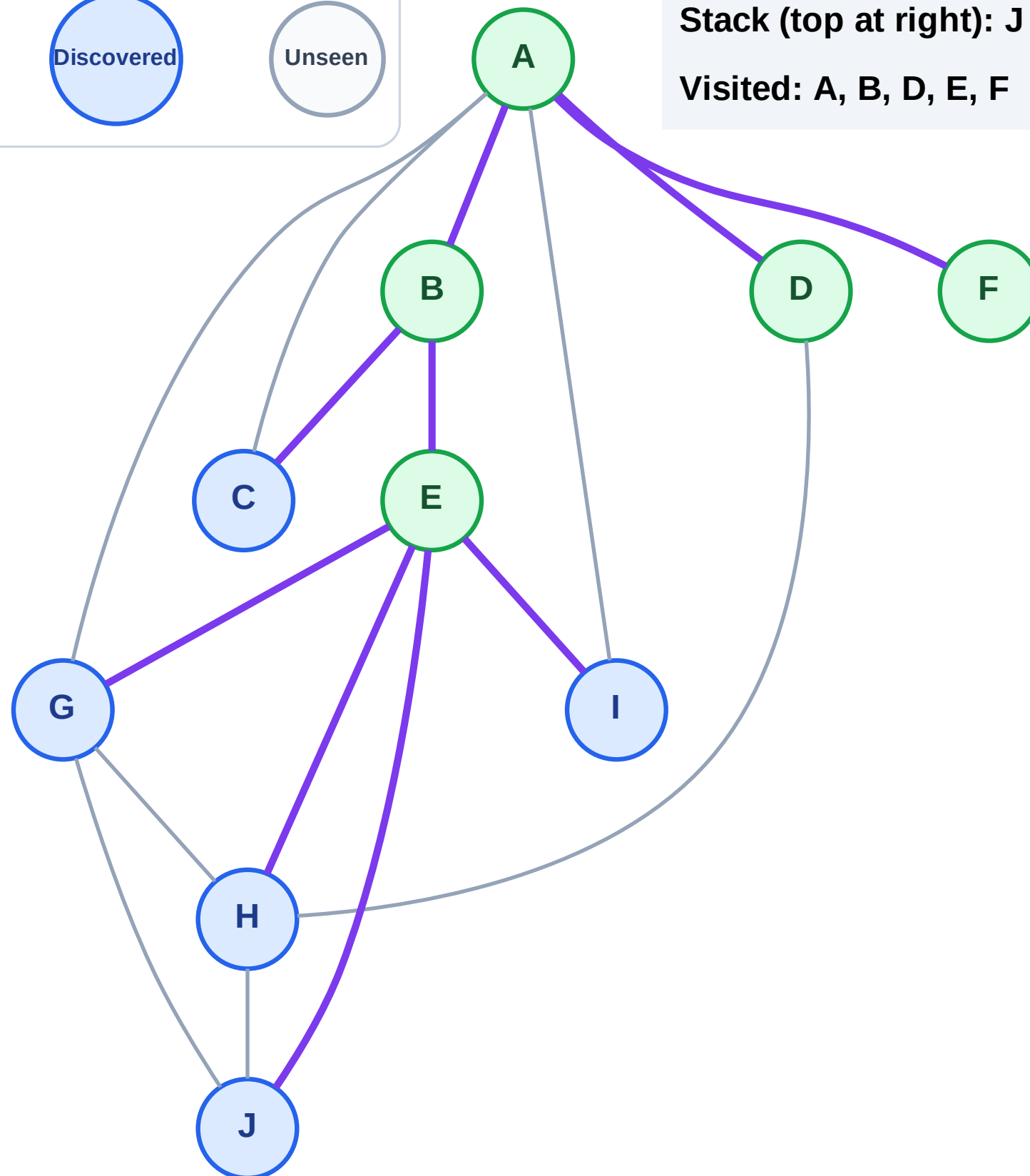
Step 11: No new neighbors from F

Legend



Stack (top at right): J | I | H | G | C

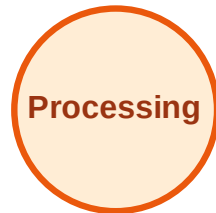
Visited: A, B, D, E, F



DFS Traversal

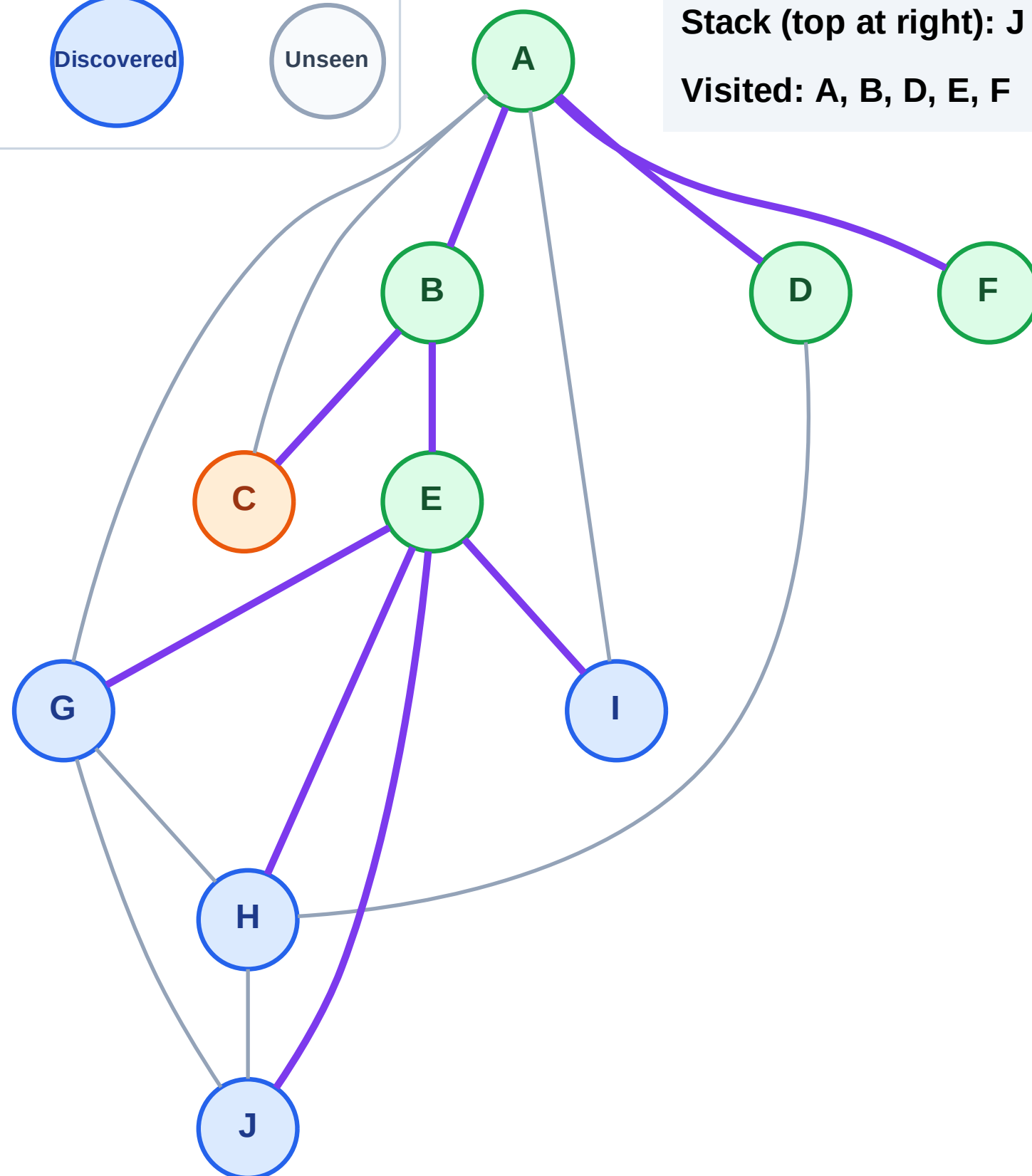
Step 12: Process node C

Legend



Stack (top at right): J | I | H | G

Visited: A, B, D, E, F



DFS Traversal

Step 13: No new neighbors from C

Legend

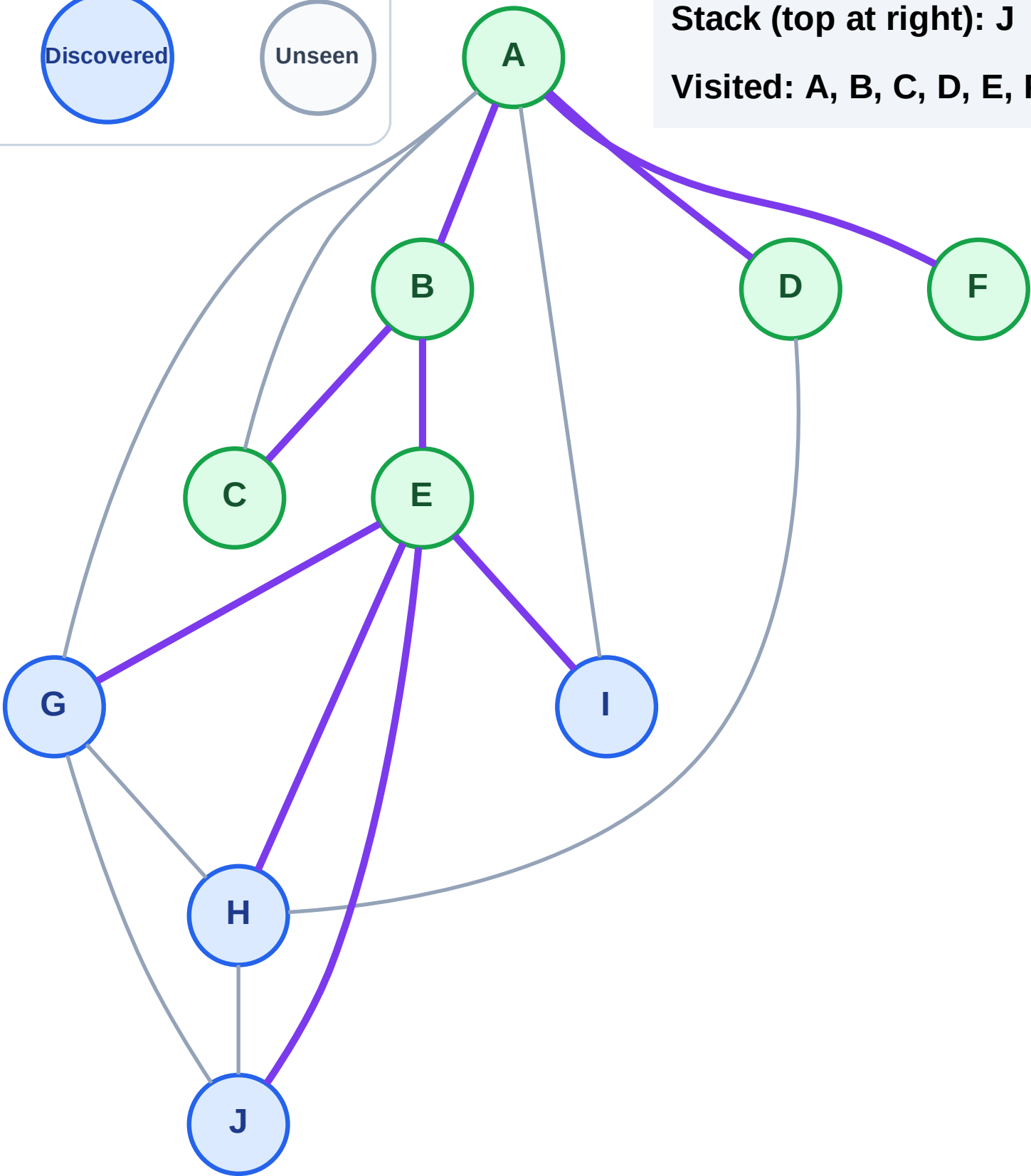
Complete

Processing

Discovered

Unseen

Stack (top at right): J | I | H | G
Visited: A, B, C, D, E, F



DFS Traversal

Step 14: Process node G

Legend

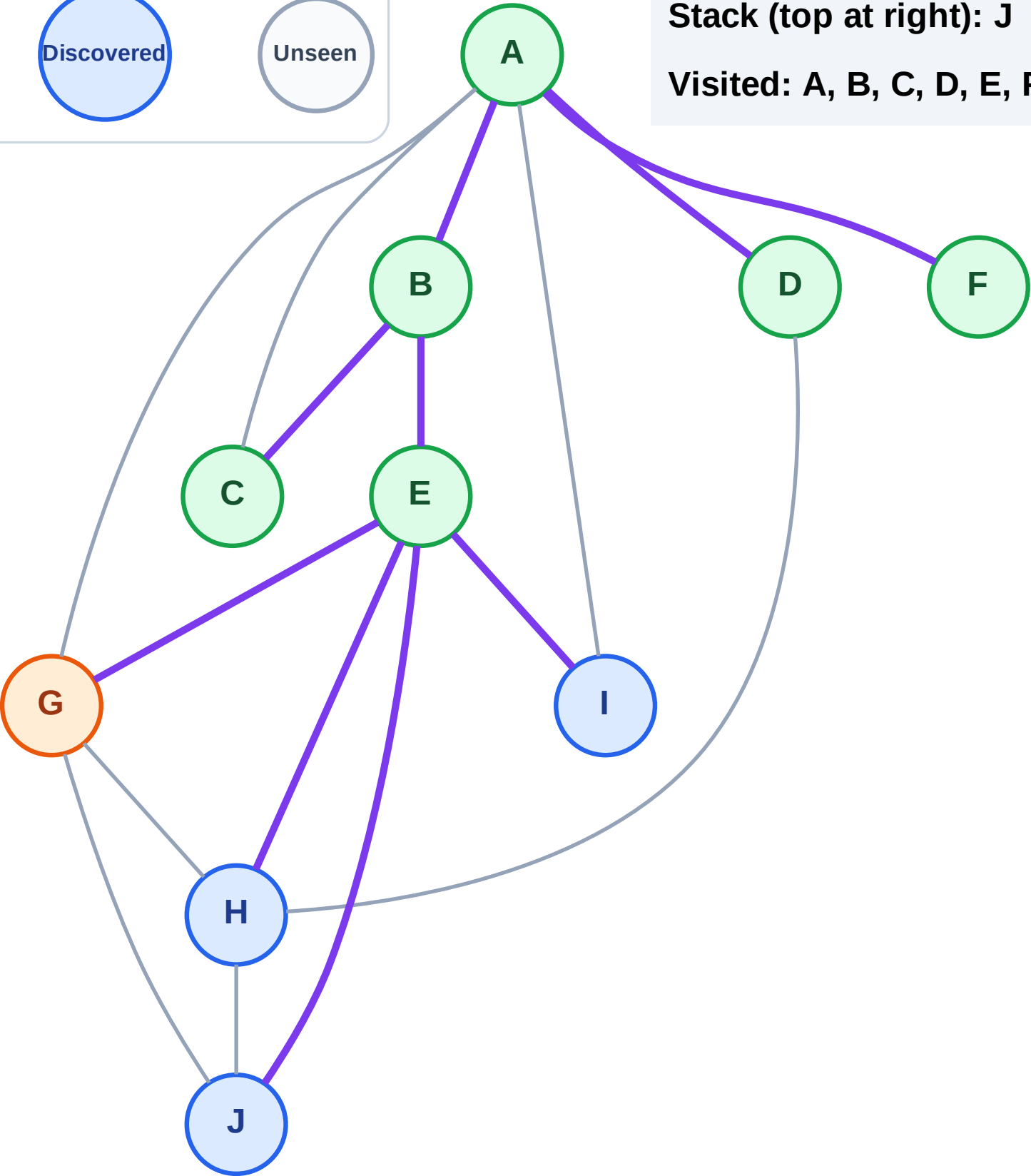
Complete

Processing

Discovered

Unseen

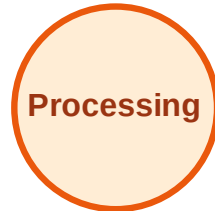
Stack (top at right): J | I | H
Visited: A, B, C, D, E, F



DFS Traversal

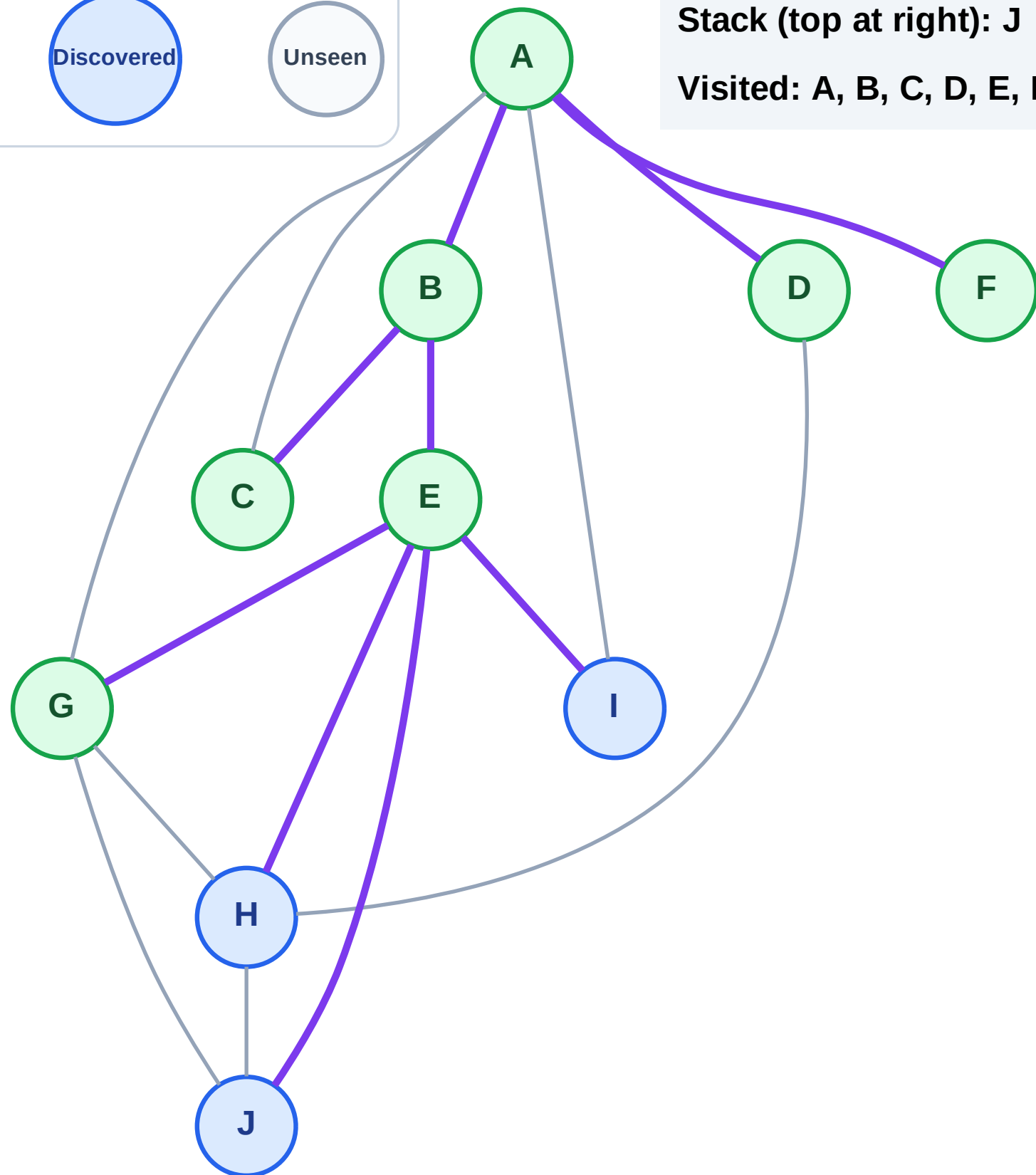
Step 15: No new neighbors from G

Legend



Stack (top at right): J | I | H

Visited: A, B, C, D, E, F, G



DFS Traversal

Step 16: Process node H

Legend

Complete

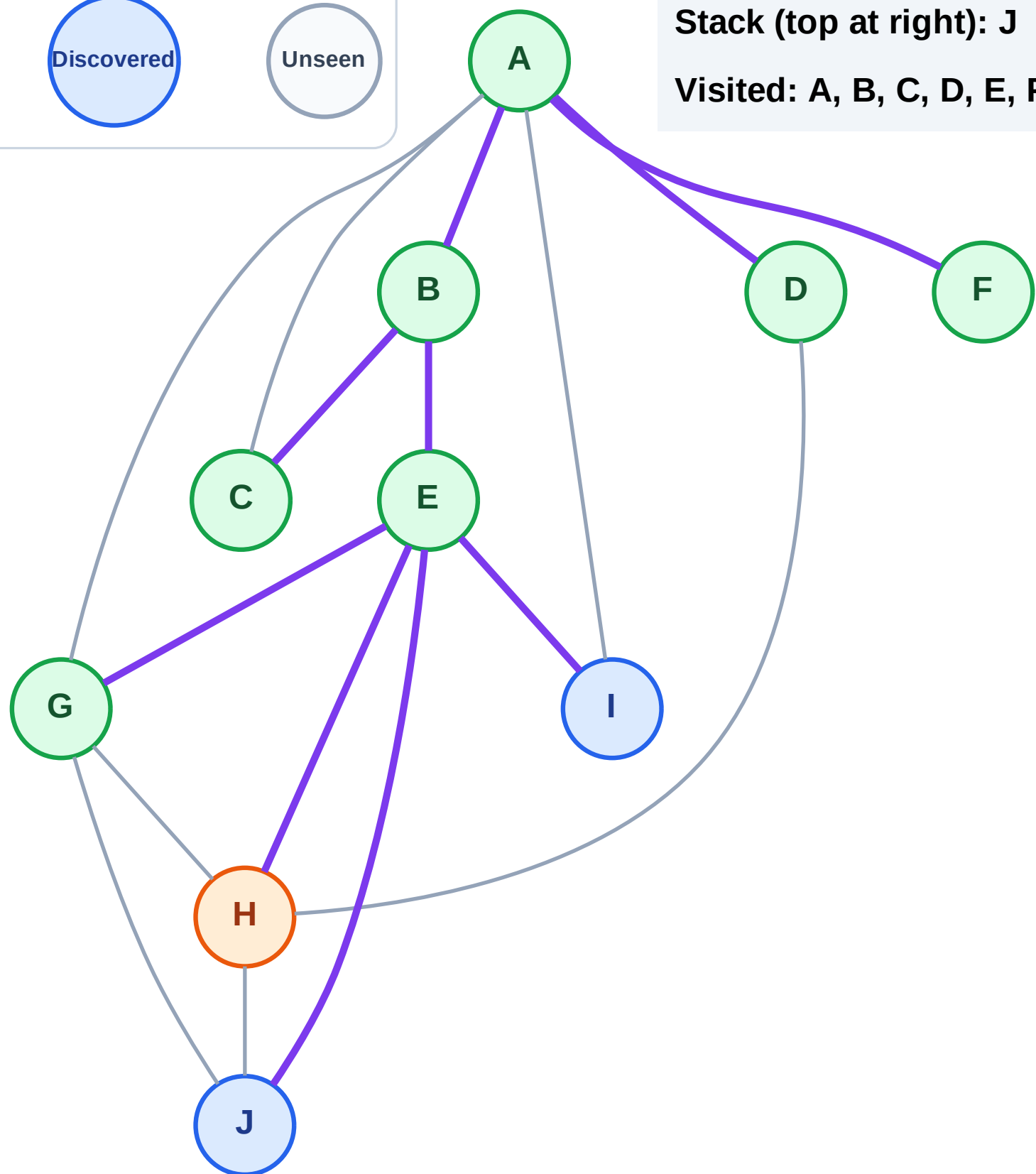
Processing

Discovered

Unseen

Stack (top at right): J | I

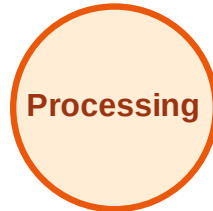
Visited: A, B, C, D, E, F, G



DFS Traversal

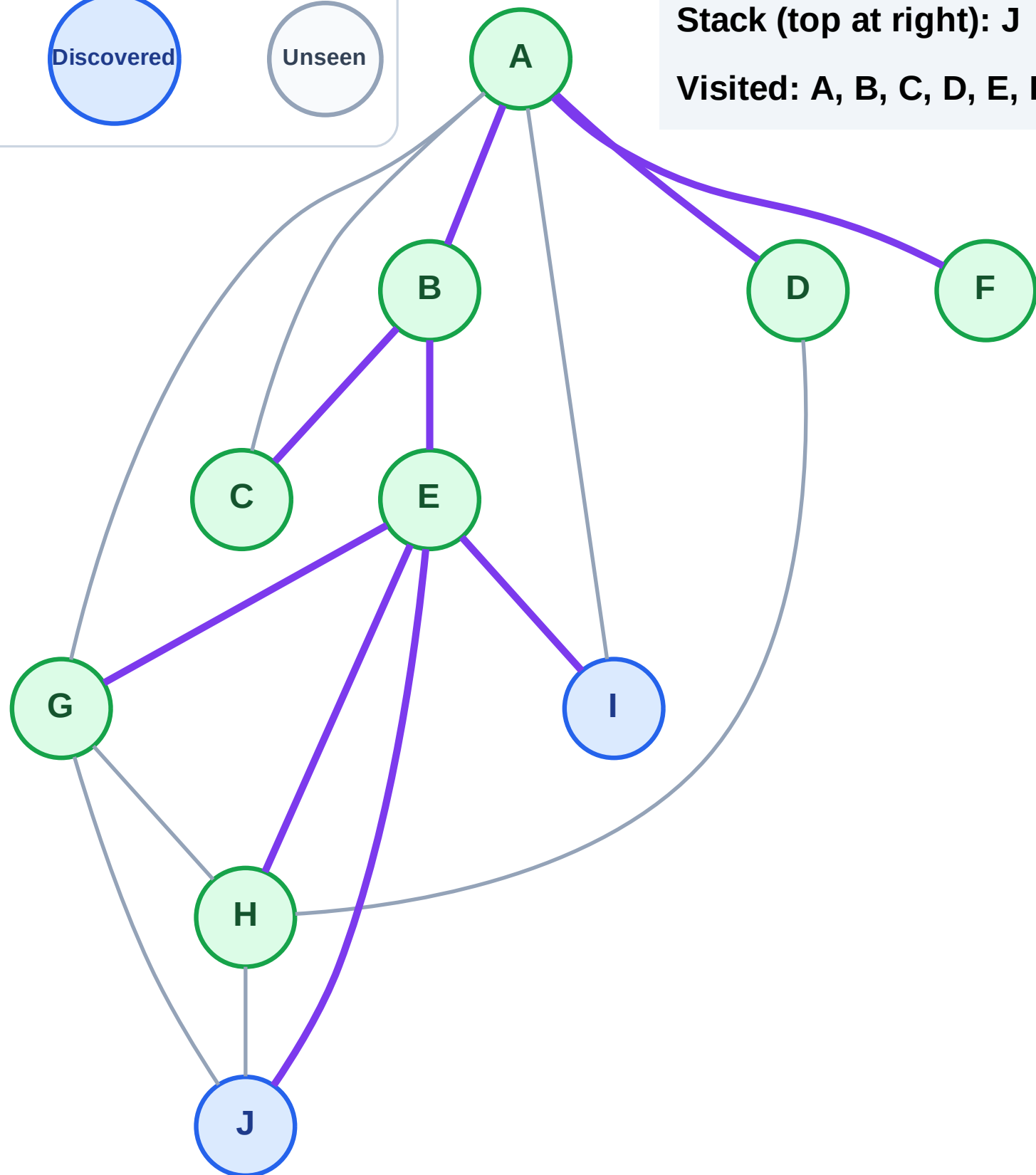
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

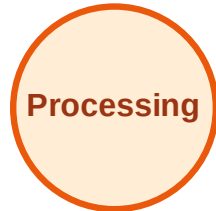
Visited: A, B, C, D, E, F, G, H



DFS Traversal

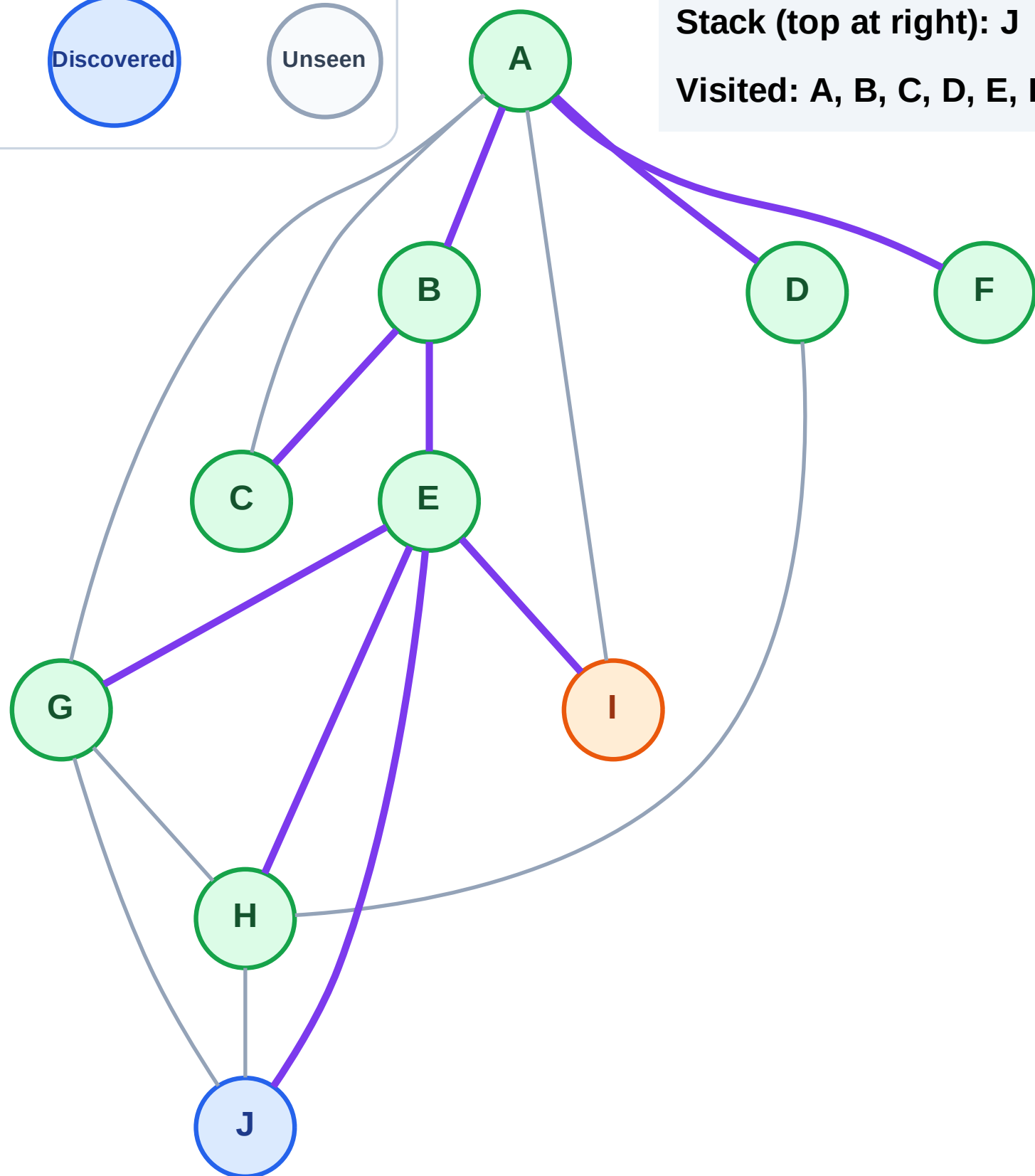
Step 18: Process node I

Legend



Stack (top at right): J

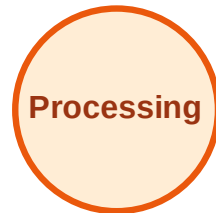
Visited: A, B, C, D, E, F, G, H



DFS Traversal

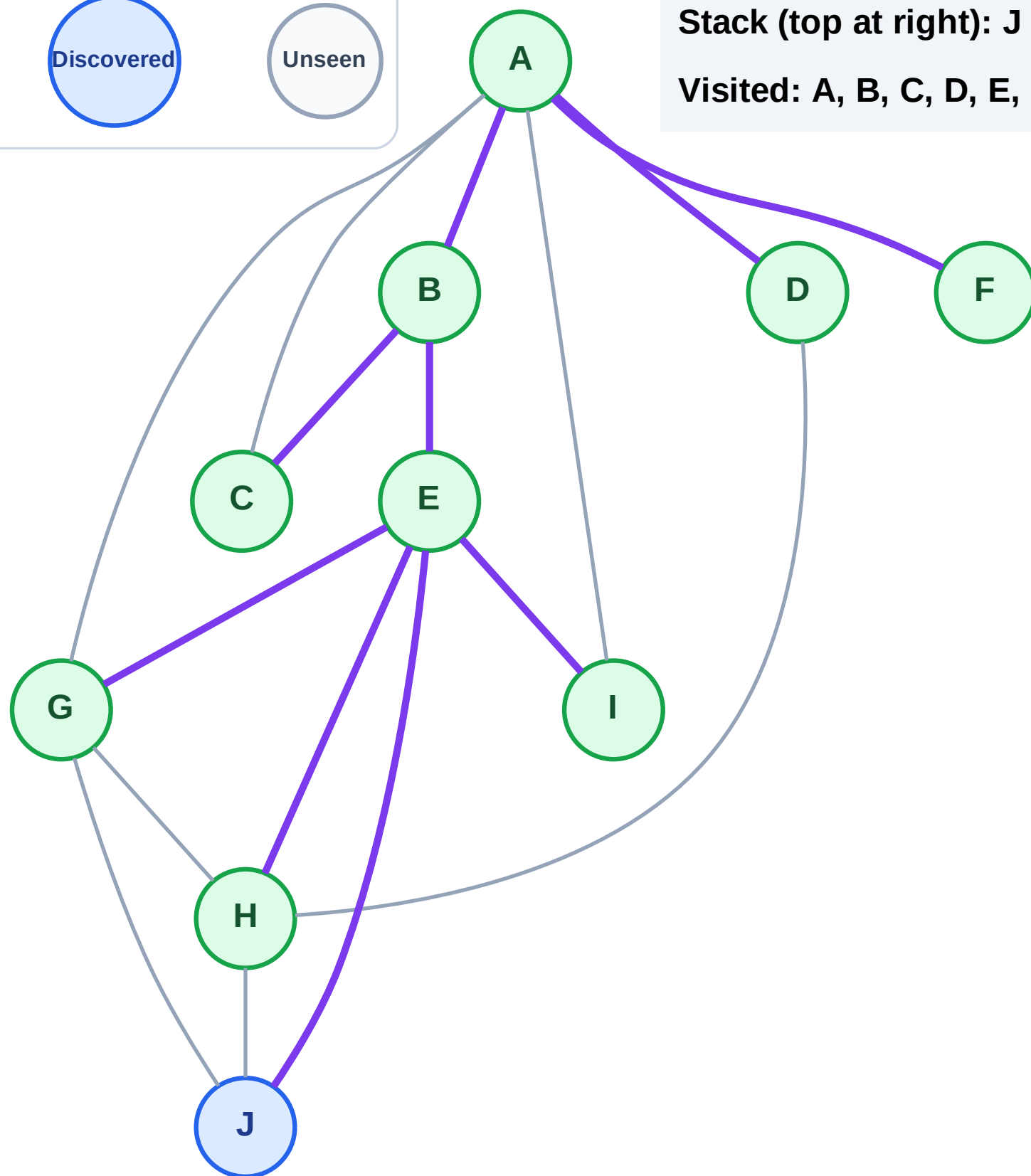
Step 19: No new neighbors from I

Legend



Stack (top at right): J

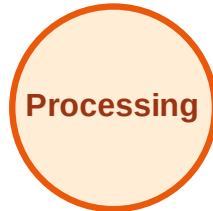
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

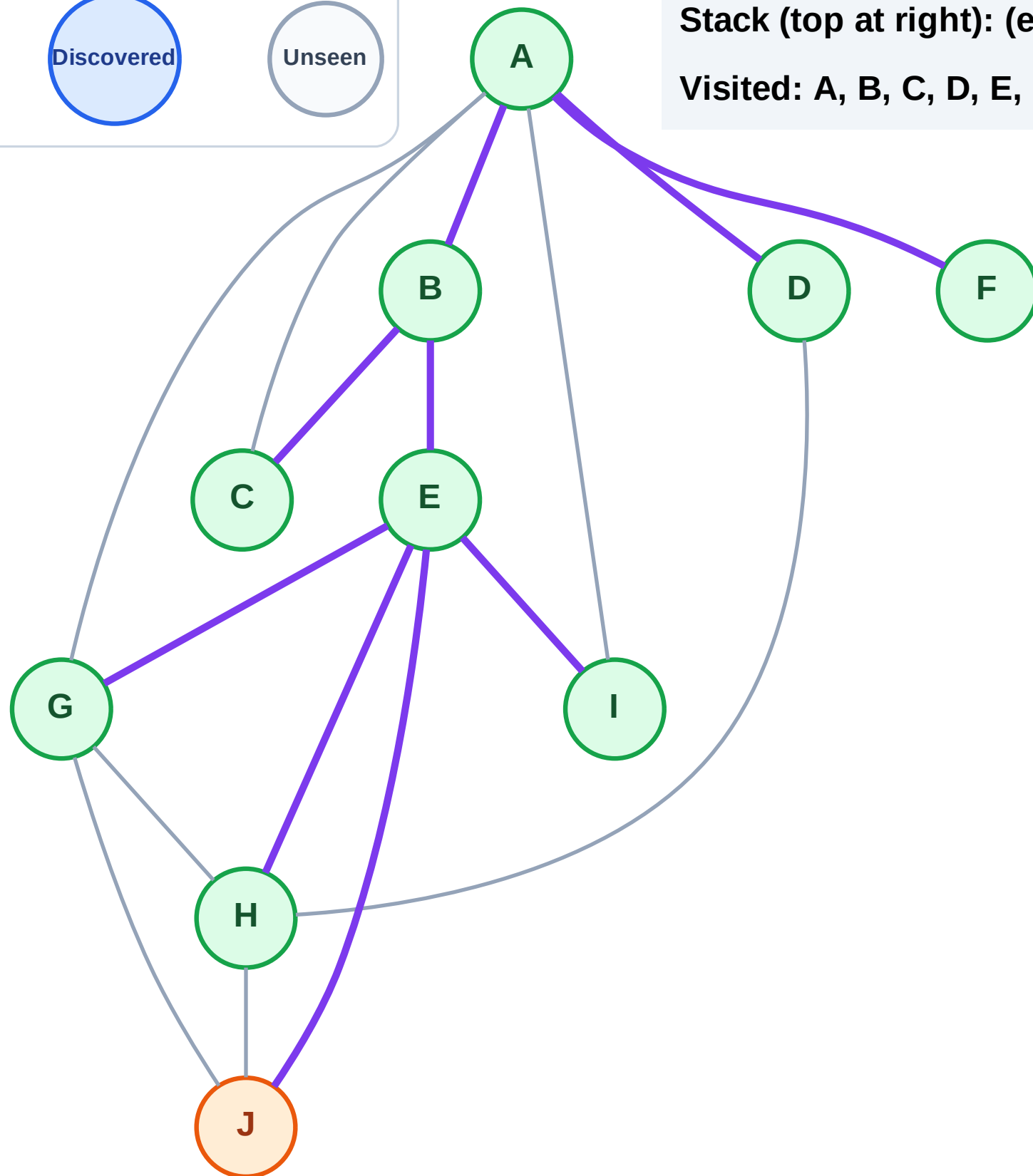
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J

